

Transparencies in
Cylinder 6626/1.

TENEMENT: Not Related.

TENEMENT HOLDER: Mt Gunson Mines Pty. Ltd.

REPORTS: Drill Logs - Sweet Nell Mine, Fair Nell Mine (Pacminex 1970) 3-9-XP Series Where X = 1 to 22 - Woocalla - Ironstone Lagoon. Pgs. 3 - 18

Drill Logs - Woocalla Area (Pacminex 1971) 3-11-XP Series where X = 1 to 9. Pgs. 19 - 37

Drill Logs - Monalena Area (Woocalla) (Pacminex 1972) Mona XP Series where X = 1 to 16. Pgs. 38 - 77

PLANS: Sweet Nell Mines - Drillhole Location (Drg No. 94). 6626-1
Sweet Nell Mines - Geological Cross Sections (Drg No 104). 6626-2
Woocalla - Ironstone Lagoon Area Geology and Drillhole Locations (Drg No. 152). 6626-3
Sweet Nell Mines - Drillhole Locations (Drg No. 94A). 6626-1(T)
Sweet Nell Mines Geological Cross Sections (Drg No. 104A). 6626-2(T)
Sketch of Proposed Drill Traverse near Woocalla Dolomite Quarries (Drg No. 146). 6626-3(T)
Woocalla - Ironstone Lagoon Area Geology (Drg No. 152). 6626-4(T)
Fair Nell Area - Cross Sections (Drg No. 310). 6626-5(T)
Fair Nell Area - Drillhole Locations (Drg No. 312). 6626-6(T)
Woocalla Cross Sections (Along Drill Traverses to N.E. of Quarries) (Drg No. 314). 6626-7(T)
Monalena - Southern Pernatty Lagoon Area - Geology Drg No. 383). 6626-8(T)
Ironstone Lagoon - Geochemical Mud Sampling (Drg No. 393). 6626-9(T)

REPORT: Mt. Gunson, Pernatty Lagoon Geological Mapping Reports. Pgs. 78-115

PLANS: S.M.L. 527 House To Manganese Workings - Drainage 6626-4
Sampling.
Geological Maps (1) 6626-5
(2) 6626-6
(3) 6626-7

000003
HOLE No. 3-9-1
PAGE 1 OF 1

PROJECT No. 401

COMPLETED: 19-12-70

CO-ORDS: X: Y:

TOTAL DEPTH: 12.2 m.

DRILLER:

TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 100	SAMPLE No.	Assayed length	ASSAY VALUE M.G.M.				
							P.P.M.		P.P.M.		
							Cu	Pb	Zn	Ag	Fe%
0		CAINOZOIC 0 - 0.6 m	/	0							
		TAPLEY HILL FM. 0.6 - 6.4 m		0.6	NONE	0	-	-	-	-	
1		0.6 - 4.3 : Woocalla Dolomite Member		1.5	40,000		<2	10	<2	2 0.53%	
2			/		1		300	110	60	2 0.25%	
3			/	3.0	2		-	-	-	-	
4		4.3 - 6.4 : Shale	/	4.3	3	5	110	60	<2	<2 1.23%	
5			/	4.9	4	5	1.75%	770	2500	16 2.4%	
6			/	5.5	5	5	3600	1030	4120	6 1.35%	
7		PANDURRA FM. 6.4 - 12.2 m	/	6.1	6	5	300	50	160	<2 0.37%	
		6.4 - 7.9 : Red Quartzite	/	6.7	7		-	-	-	-	
8		7.9 - 10.7 : Brick red quartzite	/	7.3	8		-	-	-	-	
9			/	7.9	9		-	-	-	-	
10			/	9.1	10		-	-	-	-	
11		10.7 - 12.2 : Red Quartzite	/	10.7	40,011		-	-	-	-	
12		END	/	12.2							

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HOLE No 3-9-2P

PAGE 1 OF 1

PROJECT AREA:	PROJECT No. 401	
SPUDDED:	COMPLETED: 19-12-70	
LOCATION: SW66T NE22	CO-ORDS: X: Y:	
ELEVATION: 65.6 m	DIRECTION:	INCLINATION: TOTAL DEPTH: 12.2 m
HOLE TYPE: AIRTRAC	DRILLING CONTRACTOR:	DRILLER:
LOGGED BY: P.C.	DATE:	BIT SIZE: TO TO TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE M.G.M.				
							P.P.M.		P.P.M.		% Fe
							Cu	Pb	Zn	Ag	
	0	CANON201C 0-0.6 m									
	1	TAPLEY HILL FM. 0.6-7.0 m	/	0.6							
		0.6-4.6: Woocalla Dolomite	/	1.5	40,012		-	-	-	-	
	2		/								
			/		13		110	<10	70	<1	0.46
	3		/	3.0							
			/		14		180	40	80	<1	0.45
	4		/	4.6							
		4.6-7.0: Shale			15		1400	340	290	1	1.29
	5			5.2							
				5.8	16		1300	140	220	<1	1.23
	6			6.1	17		1.07%	120	270	1	1.75
				6.7	18		8000	270	500	<1	3.80
	7	PANDURRA FM. 7.0-12.2 m		7.3	40,019 40,020		250	10	10	7	0.4
	8	Red Quartzite			21		-	-	-	-	-
	9			9.1							
	10				22		-	-	-	-	-
	11			10.7							
					40,023		-	-	-	-	-
	12	END		12.2							

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HOLE No 3-9-4P

PAGE 1 OF 1

PROJECT AREA:	PROJECT No.	4601
SUDDDED:	COMPLETED:	19-12-70
LOCATION: SWEET WELL	CO-ORDS: X:	Y:
ELEVATION: 65.4 m	DIRECTION:	INCLINATION:
HOLE TYPE: AIRTRAC	DRILLING CONTRACTOR:	TOTAL DEPTH: 7.6 m
LOGGED BY: P.C.	DATE:	DRILLER:
BIT SIZE:	TO	TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE M.G.M.				
							PPM		PPM		% Fe
							Cu	Pb	Zn	Ag	
	0	CAINOZOIC 0-0.6m		0.6							
	1	TAPLEY HILL FM. 0.6-5.2m	/	1.2	40033		140	<10	80	<1	0.44
		0.6-3.7: Woocalla Dolomite	/								
	2		/		34		-	-	-	-	-
	3		/	3.0	35		600	-	-	-	-
	4	3.7-5.2: Shale	/	3.7	36		360	130	180	<1	2.0
			/	4.3	37		400	-	-	-	-
	5	PANDURRA FM. 5.2-7.6m	/	4.9	38		200	-	-	-	-
	6		/	5.5	39		-	-	-	-	-
		Red quartzite	/	6.1			-	-	-	-	-
	7		/		40040		-	-	-	-	-
		END	/	7.6							

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0000008
HOLE No. 3-9-6 P
PAGE 1 OF 1

PROJECT AREA:

PROJECT No. 401

SPUDED:

COMPLETED: 20-12-70

LOCATION: SWEET WELL

CO-ORDS: X: Y:

ELEVATION: 65.3 m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 13.7 m

HOLE TYPE: AIRTRAC

DRILLING CONTRACTOR:

DRILLER:

LOGGED BY: P.C.

DATE:

BIT SIZE:

TO

TO

TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE MGM				
							PPM		PPM		% Fe
							Cu	Pb	Zn	Ag	
	0	CAINOZOIC 0 - 0.6 m									
	1	TAPLEY HILL Fm. 0.6 - 11.9 m		0.6							
	2	0.6 - 8.5 m: Wooralla Dolomite		1.5	40,057		90	20	80	1	0.48
	3				58		200	-	-	2	-
	4				59		200	-	-	1	-
	5			4.6	60		300	80	30	<1	0.40
	6			6.1	61		200	-	-	1	-
	7			7.6	62		300	100	50	2	0.68
	8	8.5 - 11.9 m: Shale		9.1	63		200	90	65	<1	1.22
	9			9.7	64		7100	150	430	8	1.21
	10			10.4	65		4400	150	285	5	1.95
	11			11.0	66		5600	580	1800	7	2.5
	12	PANDURRA Fm. 11.9 - 13.7 m		11.6	67		700	-	-	<1	-
	13	Red Quartzite		12.2	40,068		-	-	-	-	-
		END		13.7							

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HOLE No 3-9-72

PAGE 1 OF 1

PROJECT AREA:	PROJECT No. 401		
SPUDED:	COMPLETED: 20-12-70		
LOCATION: SWEET NELL	CO-ORDS: X:		Y:
ELEVATION: 67.2 m	DIRECTION:	INCLINATION:	TOTAL DEPTH: 11 m
HOLE TYPE: AIRTRAC	DRILLING CONTRACTOR:		DRILLER:
LOGGED BY: P.C.	DATE:	BIT SIZE:	TO TO TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:	SAMPLE No.	Assayed length	ASSAY VALUE M.G.M.				
							ppm		ppm		%
							Cu	Pb	Zn	Ag	
	0	CAINODOLIC 0-0.6		0.6							
	1	TAPLEY HILL FM. 0.6-8.8		1.5	40,069		190	20	40	<1	0.36
	2	0.6-6.1: Woscalla Dolomite			70		400	100	80		0.35
	3			3.0							
	4			4.6	71		600	80	60	1	0.89
	5				72		500	150	100	<1	0.47
	6	6.1-8.8: shale		6.1							
	7			7.3	73		1100	350	180	5	0.30
	8			7.9	74		2600	60	170	<1	1.56
	9			8.5	75		2450	900	5000	29	3.40
	9	PANDURRA FM. 8.8-11.0 m.		9.1	76		300	90	140	1	0.35
	10	Red quartzite to 9.1 m then Brick red quartzite			40,077		-	-	-	-	-
	11	END		11.0							

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TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 100	SAMPLE No.	Assayed length	ASSAY VALUE M.G.M.				
							ppm		ppm.		%
							Cu	Pb	Zn	Ag	
	0	CAINOZOIC 0 - 0.6									
	1	TAPLEY HILL Fm. 0.6 - 8.8 m	/	0.6							
			/		40,078		160	<10	10	3	0.44
	2	0.6 - 5.2: Woocalla Dolomite	/	1.8							
			/		79		-	-	-	-	
	3		/	3.0							
			/		80		-	-	-	-	
	4		/	4.6							
	5	5.2 - 8.8: Shale	/	5.2	81		400	100	70	1	1.15
			/	5.8	82		600	100	130	4	1.50
	6		/		83		700	90	139	<1	1.64
			/	6.7							
	7		/	7.3	84		1.20%	220	400	18	4.32
			/		85		1.0%	505	2000	9	6.70
	8		/	7.9							
			/	8.5	86		1600	150	500	3	1.28
	9	PANDURRA Fm.	/	9.1	87		800	80	150	<1	0.57
	10	Red quartzite	/		40,088		-	-	-	-	
			/	10.7							
		END									

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HOLE No 3-9-18P
PAGE / OF /

PROJECT AREA: Gunson

PROJECT No. 4601

SPUDED:

COMPLETED: 21.12.70

LOCATION: FAIR NELL

CO-ORDS: X: Y:

ELEVATION: 67m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 12.2m

HOLE TYPE: AIRTRAC

DRILLING CONTRACTOR:

DRILLER:

LOGGED BY: M. SLEZIAK

DATE:

BIT SIZE:

TO

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CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE				
							MGM				
							ppm		ppm		%
							Cu	Pb	Zn	Ag	Fe
	0	CALN02015 0 - 0.6									
	1	TAPLEY HILL FM. 0.6 - 10.7m	/	0.6	40,130		10	30	50	<1	0.50
	2	0.6 - 7.9 : Woocalla Dolomite	/	1.5							
	3		/		31		-	-	-	-	-
	4		/	3.0							
	5		/		32		-	-	-	-	-
	6		/	4.6							
	7		/		33		-	-	-	-	-
	8		/	6.1							
	9	7.9 - 10.7 : Shale (Black-grey clay)	/	7.6	34		-	-	-	-	-
	10		/		35	1.5	2300	90	300	<1	1.40
	11	PANDJRAA FM. 10.7 - 12.2m	/	9.1	36	1.5	2400	450	670	13	2.50
	12		/	10.7	40,137	1.5	-	-	-	-	-
		END		12.2							

TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 100	SAMPLE No.	Assayed length	ASSAY VALUE				
							MGM				
							ppm Cu	Pb	ppm Zn	Ag	% Fe
0		CAINUZOIC 0 - 0.9									
1		TADLEY HILL FM. 0.9 - 9.7 m	/	0.9							
			/	1.5	40,138	0.6	2100	<10	910	2 0.60	
2		0.9 - 4.9 : Woodcote Dolomite	/								
			/		39	1.5	-	-	-	-	
3			/	3.0							
			/								
4			/		40	1.6	-	-	-	-	
			/								
5		4.9 - 9.7 : Shale (Black grey clay)	/	4.6							
					41	1.5	800	70	1000	1 1.36	
6				6.1							
					42	1.5	3000	90	1300	<1 1.8	
7				7.6							
					43	1.5	5400	1100	930	11 2.52	
9				9.1							
10		PANDJARRA FM. 9.7 - 10.7 m			40,144	1.6	600	130	260	2 0.43	
		END		10.7							

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CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE MG/M				
							ppm		ppm		%
							Cu	Pb	Zn	Ag	Fe
	0	CAINOZOIC 0 - 1.2 m									
	1	TAPLEY HILL FM. 1.2 - 9.7 m									
	2	1.2 - 6.7 : Woocalla Dolomite	/	1.2	40,163		53	20	40	1 0.61	
			/	1.5			-	-	-	-	
	3		/		64		-	-	-	-	
	4		/	3.0			-	-	-	-	
	5		/		65		-	-	-	-	
	6		/	4.6			300	70	172	1 1.46	
	7	6.7 - 9.7 : Shale	/	6.1	66		1300	190	1710	1 8.05	
	8		/	7.6	67		2600	350	1030	9 3.10	
	9		/	9.1			1000	150	550	3 1.74	
	10	PANDURRA FM. 9.7 - 10.7 m	/	10.7	40,169						
		END	/								
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PROJECT AREA: GUNSON

PROJECT No. 401

SPUDDER:

COMPLETED: 21. 12. 70

LOCATION: FAIR WELK

CO-ORDS: X: Y:

ELEVATION: 67 m. DIRECTION:

INCLINATION:

TOTAL DEPTH: 9.1 m

HOLE TYPE: AIRTAC

DRILLING CONTRACTOR:

DRILLER:

LOGGED BY: SLB2/ALH DATE:

BIT SIZE:

TO

TO

TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE mgm				
							ppm		ppm		% Fe
							Cu	Pb	Zn	Ag	
	0	CAIN 0201C 0 - 0.9 m									
	1	TAPLEY HILL FM. 0.9 - 7.0 m		0.9	40,145		200	30	190	2	0.46
	2	0.9 - 4.3: Woodculla Dolomite		1.5							
	3				46		-	-	-	-	
	4	4.3 - 7.0: Shale		3.0							
	5			4.6	47		800	400	300	4	3.00
	6				48		1600	350	430	<1	2.08
	7	PANDURRA FM. 7.0 - 9.1 m		6.1	49		3500	250	520	<1	3.51
	8			7.6							
	9				40,150		600	80	170	<1	0.96
		END		9.1							

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TO

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PACMINEX PTY. LTD.

HOLE No. 3-11-1 P

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PROJECT AREA: GUNSON PROJECT No. 831
SPUDED: 13. 7. 71 COMPLETED: 13. 7. 71
LOCATION: Woocalla CO-ORDS: X: Y:
ELEVATION: DIRECTION: INCLINATION: $\checkmark$ TOTAL DEPTH: 15 m
HOLE TYPE: AIRTRAC DRILLING CONTRACTOR: DRILLER:
LOGGED BY: DATE: BIT SIZE: TO TO TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	A.C.S.A., 1979 ASSAY VALUE p.p.m.				
							Cu	Pb	Zn	Mn	Fe
	0	CAINOZOIC: 0-0.75m		0							
	1	TAPLEY HILL DOLOMITE: 0.75-8.25		0.75							
	1.5			1.5	60009	0.75	60	80	50	7500	170
	2	Weathered massive dolomite									
	3	Khaki/brown colour with abundant dendritic Mn			60011	1.5	70	20	40	5800	96
	3.75			3.75	60012	0.75	80	100	60	8800	138
	4			4.5	60013	0.75	130				
	5			5.25	60014	0.75	100				
	6			6	60015	0.75	110	40	40	15%	164
	6.75			7.5	60016	0.75	70				
	7			7.5	60017	0.75	115				
	8			8.25	60018	0.75	115				
	9	McHEAY REGOLITH 8.25-15m		9	60019	0.75	50				
	10	Fractured white to light brown quartzite with yellow-brown to red-brown / limonite-hematite clay fracture filling.		10.5	60021	1.5	40	120	60	3100	3.2
	11			11.25	60022	0.75	20	40	30	400	116
	12			12							
	13			12.75	60025	0.75	20	60	100	800	4.3
	14			13.5							
	15			15	60027	1.5	20	80	70	330	3.7
		END OF HOLE									

* MAM, 1971 ASSAY

PACMINEX PTY. LTD.

HOLE No. 3 - 11 - 2 P

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PROJECT AREA:GUNSON

PROJECT No.831

SPUDDED:13.7.71

COMPLETED:13.7.71

LOCATION:Noocalla

CO-ORDS: X: Y:

ELEVATION:

DIRECTION:

INCLINATION:

TOTAL DEPTH:15 m

HOLE TYPE:AIATRAC

DRILLING CONTRACTOR:

DRILLER:

LOGGED BY:

DATE:

BIT SIZE:

TO

TO

TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 100	SAMPLE No.	Assayed length	A.C.S.A., 1979 ASSAY VALUE P.P.M.					
							Cu	Pb	Zn		Mn	Fe
	0	CAINOZOIC : 0 - 0.75 m		0								
	1	TAPLEY HIGH DOLOMITE: 0.75 - 8.25										
	2	Massive weathered light brown dolomite with dendrites + fracture coatings of Mn.		1.5	60049	1.5	110	80	80		7500	166
	3			3.0	60051	1.5	100	140	60		8100	190
	4	Less deeply weathered from 4.5 m.		4.5	60054	1.5	30	60	30		9300	118
	5			6.0	60055	0.75	90					
	6			6.75	60056	0.75	80					
	7			7.5	60057	0.75	50	100	50		15%	2.4
	8	McLEAY REGOLITH: 8.25 - 10.5 m		8.25	60058	0.75	85					
	9	Fractured red quartzite with brown limonitic clay fracture fill.		9.0	60059	0.75	115					
	10			9.75	60060	0.75	30	80	50		4600	4.8
	11	PANDURRA FM. QUARTZITE: 10.5 - 15 m.		10.5	60062	1.5	20	40	20		2200	178
	12	Light pink massive quartzite.		12.0	60064	1.5	30	60	20		1800	134
	13			13.5	60065	1.5	10	20	10		1500	82
	14											
	15	END OF HOLE		15.0								

PACMINEX PTY. LTD.

HOLE No. 3 - 11 - 31

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PROJECT AREA: Gwinson PROJECT No. 831

SPUDDED: 13. 7. 71 COMPLETED: 13. 7. 71

LOCATION: Woochalla CO-ORDS: X: Y:

ELEVATION: DIRECTION: INCLINATION: ✓ TOTAL DEPTH: 15 m

HOLE TYPE: AIRTRAC DRILLING CONTRACTOR: DRILLER:

LOGGED BY: DATE: BIT SIZE: TO TO TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	A.C.S.A., 1979 ASSAY VALUE P.P.M.					
							Cu	Pb	Zn		Mn	Fe
	0	CANV201C: 0 - 0.75 m		0								
	1	TAPLEY Hill Dolom.: 0.75 - 9.0 m		0.75								
			/	1.5	60028	0.75	90	60	80	4400	121	
	2	Massive dolomite, fractured and weathered, light brown. Some pink-red dolomite.	/									
			/		60030	1.5	160	40	110	6400	176	
	3	Abundant Mn as fracture fill and dendrites.	/	3.0								
			/									
	4	Less weathered from 3m.	/		60032	1.5	250	100	130	8800	10.6	
			/	4.5								
	5		/									
			/		60034	1.5	290	80	150	9000	3.2	
	6		/	6.0								
			/									
	7		/		60036	1.5	330	100	240	1.9%	4.8	
			/	7.5								
	8		/	* 8.25	60037	0.75	495					
			/									
	9		/	* 9.0	60038	0.75	250					
		McLEAY REGOLITH 9.0 - 15 m.	Fe	* 9.75	60039	0.75	140					
	10	9-9.75: yellow-brn. & red-brn. ferruginous clays. + quartzite?	Fe	9.75	60040	0.75	80	160	90	1900	6.8	
			Fe	10.5								
	11		Fe		60042	1.5	40	140	70	700	8.7	
	12	9.75 - 15.0: DK. reddish-pink fract. massive quartzite with yellow-brown, red-brown & brick red ferruginous clays.	Fe	12.0								
			Fe		60044	1.5	50	100	60	800	6.9	
	13		Fe	13.5								
			Fe		60046	1.5	20	20	30	250	178	
	14		Fe									
	15	END OF HOLE		15.0								

* MGM, 1971 ASSAY

HOLE No. 3-11-4⁸

PAGE / OF /

000022

000022

PROJECT AREA: Gunnison

PROJECT No. 831

SPUDDER: 13. 7. 71

COMPLETED: 13.7.71

LOCATION: WOODALLA

CO-ORDS: X: Y:

ELEVATION: DIRECTION: INCLINATION: ✓ TOTAL DEPTH: 12 m

HOLE TYPE: *AIRTRAC* DRILLING CONTRACTOR: DRILLER:

LOGGED BY: DATE: BIT SIZE: TO TO TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	A.C.S.A. ASSAY VALUE P.P.M.				
							Cu	Pb	Zn	Mn	Fe
0		CAMBROZOIC: 0 - 0.75 m		0							
1		TARKEY HILL DOLOMITE: 0.75 m - 3.5 m	1	0.75							
		Massive, weathered lt. brown	1	1.5	60125	0.75	140	80	70	1500 2.5	
2		dolomite Mn remaining + abundant	1								
		limonitic clay.	1		60127	1.5	680	120	220	8300 14.5	
3			1	3.0							
4		MCKEAY REGOLITH 3.5 - 6.25 m	1		60129	1.5	110	40	80	600 6.5	
		white & mauve fractured quartzite	0.5 Fe	4.5							
5		with abundant ochreous limonitic yellow	Fe		60131	1.5	60	40	40	500 4.5	
		clay and red-brown hematitic clay.	0.5	6.0							
6			0.5 Fe		60133	1.5	40	60	20	100 15.0	
7		PANDURRA FM. QUARTZITE 6.25 - 12 m.	1	7.5							
		Massive pinkish, red and			60135	1.5	30	40	20	250 106	
8		mauve - pink quartzite.		9.0							
9					60137	1.5	20	20	20	90 94	
10				10.5							
11					60139	1.5	10	40	10	70 42	
12		END OF HOLE		12.0							

PACMINEX PTY. LTD.

HOLE No. 3-11-5P

PAGE 1 OF 1
000022

PROJECT AREA: GUNSON

PROJECT No. 831

SPUDDER: 14. 7. 71

COMPLETED: 14. 7. 71

LOCATION: WOOCAHHA

CO-ORDS: X: Y:

ELEVATION: DIRECTION: INCLINATION: TOTAL DEPTH: 9 m

HOLE TYPE: AIRTAC DRILLING CONTRACTOR: DRILLER:

LOGGED BY:	DATE:	BIT SIZE:	TO	TO	TO
------------	-------	-----------	----	----	----

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	A.C. S.A., 1979 ASSAY VALUE P.P.M.					
							Cu	Pb	Zn		Mn	Fe
	0	CAINOZOIC 0 - 1.5 m		0								
	1	Pebble of quartzite and dolomite.	C2	0.75								
				1.5	60142	0.75	220	80	80	2900	6.7	
	2	TABLEY HILL DOLOMITE: 1.5 - 3.5 m	L									
		yellow-brn. dolomite c dendritic	/		60144	1.5	90	40	60	2800	2.2	
		Mn	/									
	3		/	3.0								
			/									
	4	McLEAY REGOLITH: 3.5 - 7.5 m.	0.5		60146	1.5	110	60	60	1200	2.9	
		Fractured pink - red quartzite	1	4.5								
	5	+ abundant yellow-brn. limonitic clay	0.5		60148	1.5	30	60	60	1000	4.8	
		+ red-brn. hematitic clay.	0.5									
	6		1.0	6.0								
			0.5		60150	1.5	30	40	50	1100	3.3	
	7		1.0	7.5								
	8	PANDURRA FM. QUARTZITE 7.5 - 9 m.	0.5		60152	1.5	20	40	30	700	2.0	
		white, pink & red-brn. quartzite.	0.5									
	9	END OF HOLE		9								

PACMINEX PTY. LTD.

HOLE No. 3-11-71

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000025

PROJECT AREA: GUNSON

PROJECT No. 831

SPUDED: 13. 7. 71

COMPLETED: 13. 7. 71

LOCATION: WOO CARRA

CO-ORDS: X: Y:

ELEVATION: DIRECTION:

INCLINATION: V

TOTAL DEPTH: 16.5 m

HOLE TYPE: AIRTRAC

DRILLING CONTRACTOR:

DRILLER:

LOGGED BY: DATE:

BIT SIZE:

TO

TO

TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	A.C.S.L., 1979 ASSAY VALUE P.P.M.					
							Cu	Pb	Zn		Mn	Fe
	0	CAINOZOIC : 0 - 0.75 m		0								
	1	TAPLEY HILL FM. 0.75 - 11.5 m		0.75								
		0.75 - 11.5 : Massive Dolomite		1.5	60085	0.75	20	80	30	6600	114	
	2											
		Khaki, weathered dolomite			60087	1.5	20	80	30	7300	106	
	3	with Mn segregations into dendrites, veins etc.		3.0								
					60089	1.5	10	60	20	7800	144	
	4	From 4.5 m: Mn % decreases		4.5								
	5				60091	1.5	10	60	30	7500	84	
	6			6.0								
					60093	1.5	10	60	40	7700	94	
	7			7.5								
					60095	1.5	20	40	70	8400	88	
	8											
	9			9.0								
	10	From 11.25 m: chips of hard black metallic mineral abundant = Mn? or FeO		9.75								
					60097	0.75	50	50	100	1.4%	114	
	11	11.5 - 12.25 : Brick red, hematitic shale, Soft, weathered.		10.5	* 60098	0.75	400					
				11.25								
	12	McLEAY REGOLITH : 11.5 - 15.0 m		12.0	60099	0.75	650	60	290	1.6%	19.0	
		Sandstone of med. - coarse qtz. grains in abundant matrix of pink to brick red hematitic clay (ochre).		12.75	60100	0.75	230	60	260	800	4.4	
	13			13.5	* 60101	0.75	40					
	14				60103	1.5	60	40	50	3600	8.4	
	15	PANDURRA FM. 15.0 - 16.5 m		15.0								
		White, siliceous quartzite & purple qtzite banding. Hard.			60105	1.5	40	40	30	400	136	
	16.5	END OF HOLE		16.5								

* MAM, 1971 ASSAY.

HOLE No. 3-11-1P
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CSR/0035

000029

PAGE 1 OF 1

CSR/0035

PACMINEX PTY. LTD.—FIELD DRILL LOG

000039

HOLE No. 3-11-3P

PAGE 1 OF 1

PROJECT: DATE STARTED: TYPE OF DRILL:
 AREA: COORDINATES: HOLE SIZE:
 LOCATION: CONTRACTOR:
 STARTED: ELEVATION: DRILLER:
 COMPLETED: DEPTH: LOGGED BY:

Metres	DESCRIPTION		SAMPLE No.	Assay Length	ASSAYS					
0.75	Lgt. brn. fractured & weathered dolomite. Abund. pinkish-red Fe-stained dol. & abund. blk. Mn as dendrites & as fract. plane coatings.	1 1	60028	0.75						
1.5	As above & greyish wht. dol.	1 1	30	1.5						
3.0	Lgt. brn. weathered & less fractured dol. & less abund. Mn & Fe-stained dol. Also yellow-brn. limonitic & red-brn. hematitic clay.	1 1	32	"						
4.5	Lgt. brn. weathered sl. fractured dol. & Mn as fracture coatings. Minor Fe-stained dol. & minor yellow & brn. ferrug. clay. Also minor wht. clay.	1 1	34	"						
6.0	As above & only trace of clay.	1 1	36	"						
7.5			9.0m							
9.75	Possibly predom. yellow-brn. & red-brn. mottled ferrug. clays & massive pinkish red g'te. Minor dol. Fullback.	1 1	9.75m 40	0.75						
10.5	Dker. reddish-pink massive g'te. & lgt. yellow-brn, red-brn. & brick-red to ferrug. clays.	1 1	42	1.5						
12.0	Dker. red to red-pink fractured g'te. & abund. red-brn. & yellow-brn. clays.	1 1	44	"						
13.5	Dk. red & pink-red less fractured g'te. & lgt. yellow-brn. & red-brn. to brick red. ferrug. clays.	1 1	46	"						
15										

000031

PAGE 1 OF 1

LOGGED BY: *GLA*

CSR/0035

000032

PAGE 1 OF 1

TYPE OF DRILL:

HOLE SIZE:

CONTRACTOR:

DRILLER:

LOGGED BY: GRF

[illegible]

PACMINEX PTY. LTD.—FIELD DRILL LOG

000033

HOLE No. 3/11-6P

PAGE 1 OF 1

PROJECT: DATE STARTED: TYPE OF DRILL:
 AREA: COORDINATES: HOLE SIZE:
 LOCATION: CONTRACTOR:
 STARTED: ELEVATION: DRILLER:
 COMPLETED: DEPTH: LOGGED BY:

Metres	DESCRIPTION		SAMPLE No.	Assay Length	ASSAYS					
0.75	Lgt. brn weathered fractured dol. ± Mn as dendritic growths & fract. surface coatings. Also pink fractured dol.	1 1	60066	0.75						
1.5	Lgt. brn. weathered fractured dol. ± abund Mn. as dendritic growths & fract. surface coatings. Minor pink fractured dol.	1 1	68	1.5						
3	Lgt. brn - khaki more massive but st. fractured dol ± Mn as above. Also minor greyish wht. clay.	1 1	70	"						
4.5	As above, but massive.	1 1	72	"						
6	Khaki colored massive dolomite ± Mn as dendritic growths & fract. surface coatings & minor wht. clay.	1 1	74	"						
7.5	Khaki massive dol. ± less abund. Mn	1 1	76	"						
9	Lgt. brn - Khaki massive dol. ± Mn. & ± minor pink-whit. gkls. Also minor wht. clay.	1 1	78	"						
10.5	Lgt. brn - Khaki massive & st. fractured dol ± Mn & pink-whit. gkls. & lgt. brn. clay.	1 1	80	"						
12	Weathered & fractured Wht, pinkish-whit - red-brn gkls. ± brick red clay (minor). Some rounded gk. sst. grns.	1 1	82	"						
13.5	Predom. brick-red & reddish-brn. clay ± pinkish-whit. to red gkls.	1 1	84	"						

McLeay Regolith.

PACMINEX PTY. LTD.—FIELD DRILL LOG

000034

HOLE No. 3-11-7P

PAGE 1 OF 2

PROJECT: DATE STARTED: TYPE OF DRILL:
 AREA: HOLE SIZE: *Written up*
 LOCATION: COORDINATES: CONTRACTOR:
 STARTED: ELEVATION: DRILLER:
 COMPLETED: DEPTH: LOGGED BY:

Metres	DESCRIPTION	SAMPLE No.	Assay Length	ASSAYS					
0	Weathered dolomite (khaki color) c dendritic Mn. Some pinkish red grm. Large chips - fractured.	60085							
1.5	As above. Mn as dendritic structures & along fract. planes.	60087							
3.0	As above.	89							
4.5	As above - Mn decreasing.	91							
6.0	Khaki weathered dolomite c minor Mn.	93							
7.5	As above	95							
9.0	As above	97							
10.5	Yellow brn dolomite c hard bluish blk. Mn chips (crushed to sandy blk) & marne-red clay.	99							
12.0	c. sst. Pinkish-white fine chips c red-brn clay & yellow-brn dolomite (fallback?)	100							
12.75	Pinkish wht. c.g. sst; abund brgt red hem clay, possibly as sst. matrix.	103							
15		105							

000035

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[illegible]

PACMINEX PTY. LTD.—FIELD DRILL LOG

000038

HOLE No. 3-11-8P

PAGE 1 OF 1

PROJECT: DATE STARTED: TYPE OF DRILL:
 AREA: COORDINATES: HOLE SIZE:
 LOCATION: CONTRACTOR:
 STARTED: ELEVATION: DRILLER:
 COMPLETED: DEPTH: LOGGED BY:

Metres	DESCRIPTION	SAMPLE No.	Assay Length	ASSAYS					
0.75	Lgt. brn. weathered & fractured dol. ± Mn. as dendritic growths & fract surface coatings. Also pinkish-red fractured dol.	60106	0.75						
1.5	Lgt. brn. less fractured dol. ± Mn as above & minor pinkish red fractured dol.	8	1.5						
3	Lgt. brn. dol. ± Mn as above & minor wht dol.	10	"						
4.5	Lgt. brn. - khaki massive dol. ± less abund Mn as dendritic growths & fract surface coatings	12	"						
6	Massive khaki dol. ± Mn as above & minor wht. clay.	14	"						
7.5	As above.	16	"						
9	massive lgt. brn - khaki dol ± abund. Mn as above & minor pink-whit. gte.	18	"						
10.5	Wht & pinkish wht & red sl. fractured gte ± greyish wht clay.	20	"						
12	Wht, pinkish wht & red (overall mauvish tinge) massive & sl. fractured gte. ± minor reddish-brn. clay.	22	"						
13.5	Predom. dk. red-brn (hematitic) clay ± wht, red-brn, & red-brn massive gte.	24	"						

McLeay. Regolith

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TYPE OF DRILL:

HOLE SIZE:

CONTRACTOR:

DRILLER:

LOGGED BY:

[illegible]

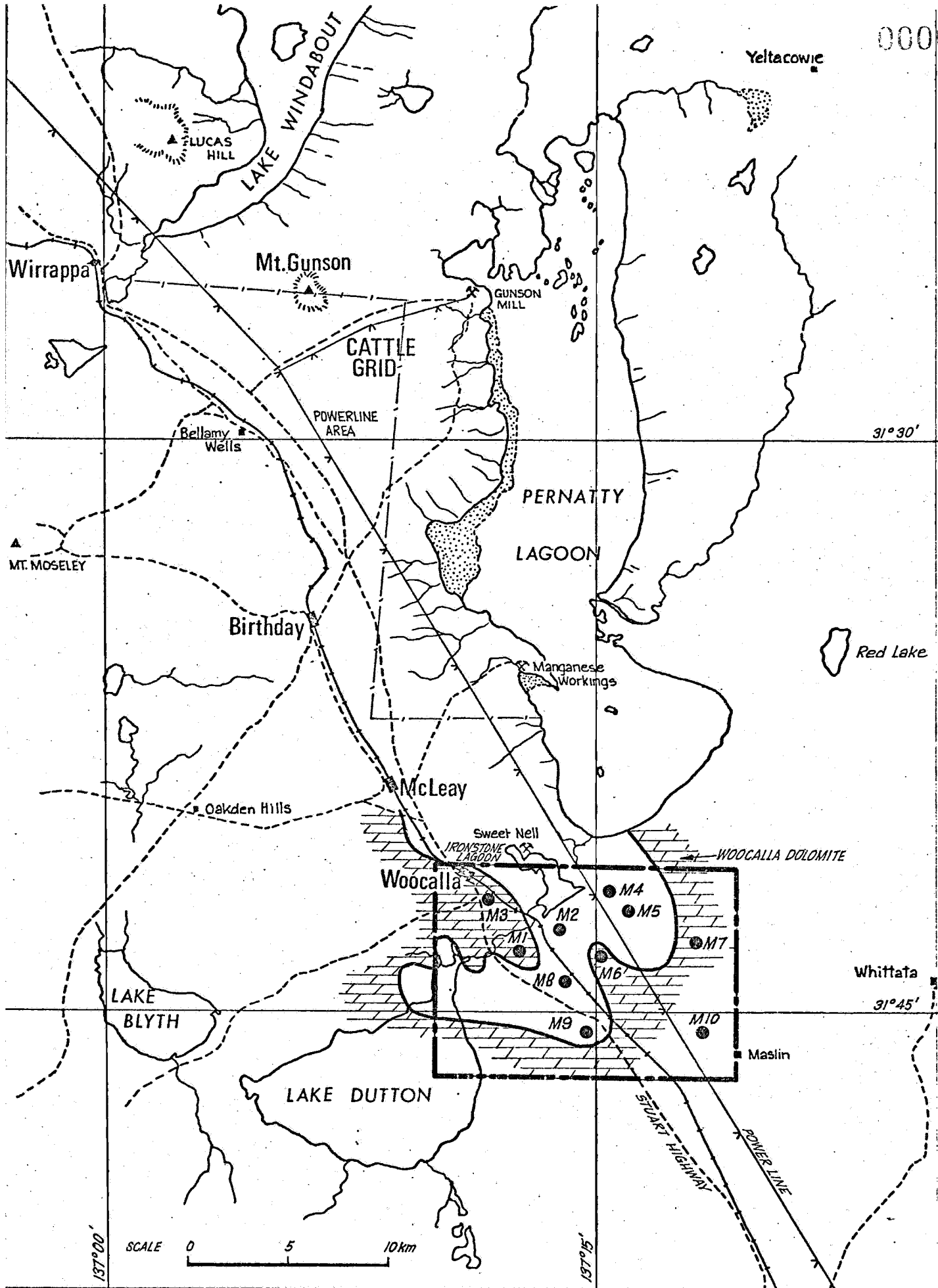


Figure 6 : Drill Hole Locations, October 1975 Drilling, Lake Dutton
INPUT Area, E.L. 199, Pernatty Lagoon, South Australia

PACMINEX PTY. LTD.

HOLE No. MONA 1P

PAGE 1 OF 1

PROJECT AREA: GUNSON PROJECT No. 401
 SPUNDED: 18/06/72 COMPLETED: 18/06/72
 LOCATION: MONALENA Map 6334 G CO-ORDS: X: 6,492,575 NY: 705,295 ME
 ELEVATION: 80m DIRECTION: INCLINATION: TOTAL DEPTH: 33 m
 HOLE TYPE: Airtrac Percussion DRILLING CONTRACTOR: O'Neill DRILLER:
 LOGGED BY: J.M.H. DATE: 1972 BIT SIZE: TO TO TO
 D.G. TONKIN 1979

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:200	SAMPLE No.	ASSAYED LENGTH	ASSAY VALUE				
							1978 by A.C.S.A.				
							P.P.m. Cu	P.P.m. Pb	% Zn	P.P.m. Fe	P.P.m. Mn
	0	CAIN/OZOLIC 0-1 m		0							
		TAPLEY HILL FORMATION 1-11.5m			69905		-	-	-	-	-
		1-4: Magazine Hill clay,			906		-	-	-	-	-
					907		25	80	50	0.6	3000
	5	4-6: Brown weathered dol.,		5	908		90	70	50	1.2	6250
					909		90	70	70	2.55	8000
		6-9: Dolomite + Mn . Light brown			910		110	90	70	2.45	1.2%
					911		45	100	40	2.75	1.18%
					912		-	-	-	-	-
	10	9-12: Dolomite Khaki + Mn, Fe-stone		10	913		35	50	50	8.1	1.26%
		red qz. pebbles 11-12m			914		60	100	80	26.5	5000
		PANDURRA FORMATION 11.5-33 m			915		40	70	80	13.5	2800
					916		40	60	80	13.0	3000
					917		60	40	75	11.0	3000
	15	wet & easy to drill.		15	918		80	40	70	8.1	2750
					919		50	60	75	8.0	3500
					920		40	40	40	6.65	2250
					921		40	20	60	7.2	2750
		11.5-26: White, fractured quartzite			922		125	70	70	6.7	2500
	20	+ yellow ochre to yellow brown clay		20	923		95	40	60	5.2	1650
		with good black heavy mineral			924		70	20	100	3.9	2250
		trace. Limonitic P.Fm.			925	*	32	20	40		
					926		30	20	40	5.35	2500
					927		90	30	60	5.0	3000
	25			25	928		30	40	45	4.5	1450
		26-33: White, fractured quartzite			929		40	20	40	4.3	1600
		+ pink-brown clay. Same H.M.			930		30	50	35	4.85	3000
		Transitional between			931		90	50	50	3.8	1300
		limonitic and hematitic.			932		120	40	65	3.5	1900
	30			30	933		35	40	40	3.1	2750
		Hole possibly drilled down			934		50	20	50	2.85	1000
		a fracture zone.			935		30	40	35	3.1	2500
	33	END OF HOLE		33	69936		40	20	35	3.1	2750

* 1972 MGM ASSAY.

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDD: 18/06/72

COMPLETED: 18/06/72

LOCATION: MONALENA Map 6334 G

CO-ORDS: X: 6,492,170 mN^Y: 705,485 mE

ELEVATION: 75 m DIRECTION:

INCLINATION:

TOTAL DEPTH: 21 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: J.M.H. DATE: 1972

BIT SIZE:

TO

TO

TO

D.G. TONKIN 1979

1979

[illegible]

D.G. TOWKIN 1979

[illegible]

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDDER: 18/06/72

COMPLETED: 19/06/72

LOCATION: MONALENA Map 6334 G

CO-ORDS: X: 6,490,594 m N Y: 705,970 m E

ELEVATION: 75 m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 45 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: J.M.H. DATE: 1972

BIT SIZE:

TO

TO

TO

+ F. Trumpelman 1972 D.G. Tonkin (1979)

D.G. Tonkin (1979)

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 200	SAMPLE NO.	ASSAYED LENGTH	ASSAY VALUE 1978 by A.C.S.A.					
							Cu	Pb	Zn	g Fe	Mn	* Co
	0	SOIL 0-2 m			0							
		WHYALLA SANDSTONE 2 - 4.5 m		Cz								
		TAPLEY HILL FORMATION 4.5-45 m			69983		15	30	20	0.65	240	
					984		20	60	30	0.90	400	
		-6: Khaki shale + dolomite			985		20	180	170	2.60	1650	20
					986		20	380	250	2.80	2750	
					987	*	41	300	470	-	-	25
					988		40	220	750	2.75	1900	
					989	*	45	150	660	-	-	15
H ₂ O	10	9: Water table			69990	*	70	170	840	-	-	20
					991		100	160	450	3.05	950	20
					992		60	160	220	3.50	1000	20
		6-45: Laminites; laminated black shales & Dark grey dolomite. Fissile,			993		50	160	210	3.50	1100	25
					994		50	120	300	3.25	1150	20
					995	*	60	130	265	-	-	25
		Shale laminae predominate over dolomite laminae			996		50	140	360	3.35	1150	25
					997	*	49	125	340	-	-	20
					998		60	160	380	3.35	1800	20
					69999		40	120	270	2.45	1750	22
	20	Samples contaminated by WSS and dune sand from 21-36 m.			55201	*	110	100	335	-	-	20
		Heaviest contamination between 30-32 m (50-75%)			202	*	35	90	310	-	-	18
					203	*	43	130	350	-	-	20
					204		55	140	400	2.8	1450	21
					205		45	120	350	2.85	1800	20
					206		35	80	260	2.9	1650	25
					207		35	90	170	2.8	1100	20
					208		40	100	290	2.4	1100	20
					209		35	80	190	2.5	1000	20
					210		60	160	600	3.75	1250	20
	30				211		40	60	220	2.75	950	15
					212		40	60	140	1.55	750	10
					213		75	100	250	2.5	1050	15
					214		100	80	350	3.05	1600	20
					215		60	120	250	3.7	1550	20
					216		35	80	340	2.65	1500	20
					217	*	50	80	360	-	-	20
					218		50	100	400	2.85	3000	25
					219	*	40	90	440	-	-	25
					220	*	43	90	445	-	-	20
	40				55221		40	120	400	2.95	5500	20
		Cont. on P.2										

PROJECT AREA:

PROJECT No.

SPUDDER:

COMPLETED:

LOCATION:

CO-ORDS: X:

Y:

ELEVATION:

DIRECTION:

INCLINATION:

TOTAL DEPTH:

45 m

HOLE TYPE:

DRILLING CONTRACTOR:

DRILLER:

LOGGED BY:

DATE:

BIT SIZE:

TO

TO

TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 200	SAMPLE No.	Assayed length	ASSAY VALUE					
							Cu	Pb	Zn	% Fe	Mn	Co
	40	TAPLEY HILL FM.(Cont.) -45 m		40	55222		30	80	550	1.56	5000	20
		Laminites; black shale & dolomite			55223		45	160	330	1.70	3250	20
					55224		30	40	190	1.3	3750	20
					55225		25	80	180	1.42	3000	12
	45	END OF HOLE		45	55226	*	58	65	250	-	-	15
						#	1972	MGm	ASSAYS			

PACMINEX PTY. LTD.

HOLE No. MONA 5P

PAGE 1 OF 2

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDDED: 19/06/72

COMPLETED: 19/06/72

LOCATION: MONALENA Map 6334 L

CO-ORDS: X: 6,489,270 N Y: 705,590 E

ELEVATION: 100 m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 45 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: D.G.T. DATE: 1979

BIT SIZE:

TO

TO

TO

F. Trumelman 1972

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:200	SAMPLE NO.	ASSAYED LENGTH	ASSAY VALUE					
							1978 by A.C.S.L.					
							Cu	Pb	Zn	% Fe	Mn	* Co
	0	CAINOZOIC 0-3										
		WHYALLA SANDSTONE 3-9½ m										
		0-4: Silicified lithic sandstone			55227		15	20	20	0.97	120	
					228		160	20	20	0.4	45	
					229		15	20	20	0.24	45	
		4-9½: Soft lithic sandstone, becoming finer grained & richer in white clay towards base			230		10	<20	10	0.23	20	
					231		15	<20	20	0.19	15	
					232		10	<20	10	0.12	20	
					233		10	40	10	0.17	20	
					234		10	<20	10	0.20	50	
	10	TAPLEY HILL FORMATION 9½-45 m			235		10	30	35	0.18	20	
		9½-26: Magazine Hill clay, sandy & white to 14m			236		15	30	10	0.17	20	
		off-white clay (brown, bluish-grey & greenish-grey) to 18 m			237		10	<20	30	0.17	20	
					238		80	20	20	0.18	50	<10
					239		45	40	10	0.37	20	
					240		25	120	20	0.43	15	
					241		25	120	20	1.36	20	65
					242		70	80	20	1.05	40	
					243		115	80	30	1.36	120	
					244		310	120	180	1.56	150	
	20	Clay: khaki to 25 m			245		90	240	500	1.76	1050	30
					246		-	-	-	-	-	-
					247		120	900	500	1.60	3500	
					248		130	600	500	1.60	2750	
					249		100	160	240	1.50	1600	
					250		40	180	360	1.90	2500	
		Becoming dark grey clay by 25 m			251		30	220	390	1.70	1300	25
		25-34: Black & Dk. grey shales			252		40	160	210	1.56	1650	25
					253		45	180	210	1.66	1800	25
					254		45	180	200	1.66	1700	
	30				255		40	140	270	1.5	1600	20
					256		35	180	240	1.46	1700	20
					257		35	140	240	1.60	2000	15
					258		55	120	300	1.46	1850	
					259		45	180	500	1.4	1750	20
		34-45: laminated dark grey shales with shale laminae dominant over light grey dolomite laminae.			260		60	120	400	1.46	3000	20
					261		115	160	310	1.30	1600	
					262		40	180	390	1.33	2000	20
					263		30	120	500	1.26	2000	
					264		40	140	550	1.28	1900	20
	40	Cont. on P.2			55265		60	160	340	1.42	1800	

PROJECT AREA:

PROJECT No.

SPUDDER:

COMPLETED:

LOCATION:

CO-ORDS: X:

Y:

ELEVATION:

DIRECTION:

INCLINATION:

TOTAL DEPTH: 45 m

45 m

HOLE TYPE:

DRILLING CONTRACTOR:

DRILLER:

LOGGED BY:

DATE:

BIT SIZE:

TO

TO

TO

[illegible]

000046

PACMINEX PTY. LTD.

HOLE No. MONA 6P

PAGE 1 OF 1

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDED: 20/06/72

COMPLETED: 20/06/72

LOCATION: MONALENA Map 6334 L

CO-ORDS: X: 6,489,085m NY: 705,790 m E

ELEVATION: 90 m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 36 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: F.T.

DATE: 1972

BIT SIZE:

TO

TO

TO

D.G. Tonkin 1979

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 200	SAMPLE NO.	ASSAYED LENGTH	ASSAY VALUE					
							1978 by A.C.S.L.					
							Cu	Pb	Zn	% Fe	Mn	* Co
	0	RECENT SEDIMENTS 0-2 m		0								
		Sand, soil & rubble of QZ, DOL, S										
		TAPLEY HILL FORMATION 2-27 m			55271		70	60	240	1.78	4500	
		2-5.5: Magazine Hill clay. Brown			272		50	100	230	1.60	7500	
		to buff + Tr. dendritic			273		140	180	210	1.50	5750	
		Fe/Mn			274		400	90	190	1.55	4750	
		5.5-27: DOLOMITE			275	*	370	190	250	-	-	85
		5.5-15: Pink dolomite			276		180	40	170	1.15	8500	
					277		180	80	230	0.96	7250	
	10			10	278		90	30	100	0.79	7000	
					279	*	130	45	115	-	-	25
					280		80	20	80	0.77	7750	
					281		50	50	60	0.83	8750	
					282		45	40	55	0.71	8250	
					283		80	30	85	0.76	8500	
		15-18: Pink dolomite + light			284	*	79	75	50	-	-	15
		bluish-grey shale. Tr. HM			285		65	20	45	0.72	8500	
					286		60	40	40	0.63	6750	
		18-22: Light brown-buff dolomite			287		70	70	55	0.79	0.94 %	
	20	+ TR. bluish shale. Tr. HM		20	288	*	45	55	45	-	-	20
					289		30	80	35	0.68	1.08 %	
					290		30	80	40	0.77	1.04 %	
		22-27: Light bluish-grey dolomite			291		10	20	30	0.68	1.22 %	
		V. minor black			292	*	40	55	40	-	-	25
		Dol. massive, tight,			293		30	60	45	0.92	1.14	
		clean; light brown-buff			294		20	20	35	0.76	1.32	
		in places			295	*	37	60	40	-	-	30
		PANDURRA FORMATION 27-36 m			296	*	28	40	35	-	-	10
					297	*	12	10	12	-	-	<10
	30	Brick red sandstone/quartzite		30	298	*	20	15	15	-	-	<10
		+ good trace of Heavy Minerals			299	*	15	25	20	-	-	<10
					55300	*	12	15	12	-	-	<10
		Possibly ? McLEAY BEDS ?			55301	*	18	10	10	-	-	<10
					302	*	18	20	20	-	-	<10
					303	*	17	10	13	-	-	<10
	36	END OF HOLE		36	304	*	18	10	15	-	-	<10

* 1972 MAM ASSAY

000047

PACMINEX PTY. LTD.

HOLE No. MONA 7P

PAGE 1 OF 1

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDDED: 20/06/72

COMPLETED: 20/06/72

LOCATION: MONALENA Map 6334 L

CO-ORDS: X: 6,489,105mN Y: 705,925mE

ELEVATION: 100 m DIRECTION:

INCLINATION:

TOTAL DEPTH: 30 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: F.T.

DATE: 1972

BIT SIZE:

TO

TO

TO

D.G. TONKIN 1979

CORE RECOVERY	METRE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:200	SAMPLE NO.	ASSAYED LENGTH	ASSAY VALUE					
							1978 by A.C.S.K.					
							Cu	Pb	Zn	% Fe	Mn	* Co
	0	RECENT SEDIMENTS 0-2.5 m		0								
		Sand, soil, clay, RQZ & DOL float			55305		20	40	30	1.34	1.30	%
		T.H. Fm. DOLOMITE 2.5-10.5m			306		20	60	35	1.13	1.42	%
		2.5-7: Dark green to khaki-grey dolomite + Fe/Mn staining			307	*	20	45	25	-	-	20
					308		30	20	30	0.77	1.46	%
					309		20	30	40	0.82	1.30	%
		7-10.5: Dark ^{green-} grey dolomite			310		45	40	30	0.80	1.32	% 25
		Mn mottling more pervasive than above.			311		45	50	30	1.03	1.46	
	10			10	312		30	30	30	0.85	1.32	
		PANDURRA FORMATION 10.5-30 m			313		30	40	30	0.75	8500	10
		10.5-17: Pink sandstone/quartzite			314		15	20	20	0.46	550	<10
					315	*	32	20	25	-	-	<10
					316		10	<20	20	0.54	600	
					317		10	30	20	0.69	50	<10
					318		20	60	20	0.69	450	<10
					319		10	20	10	0.70	450	<10
		17-20: Very dark red			320		25	<20	20	1.54	600	<10
					321		10	20	10	1.28	300	<10
	20	20-21: Brown shaley band		20	322		10	30	10	1.34	450	<10
		21-25: Pink quartzite			323	*	53	10	40	-	-	15
					324		10	40	30	0.97	350	<10
					325		20	40	20	1.03	200	<10
					326		10	20	20	1.11	300	<10
					327		10	20	20	1.17	400	<10
		25-27: Darker pink quartzite			328		5	<20	15	1.59	200	<10
					55329		10	<20	15	1.42	250	<10
		29-30: Dark red-brown quartzite			55331		5	30	20	1.13	150	<10
					332		20	20	20	1.09	350	<10
	30	END OF HOLE		30	55333		10	40	10	1.78	300	10

* 1978 MGM ASSAY

PACMINEX PTY. LTD.

HOLE No. MONA 8P

PAGE 1 OF 1

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDED: 20/06/72

COMPLETED: 20/06/72

LOCATION: MONALENA Map 6334 G (In quarry) CO-ORDS: X: 6,490,655 mNY: 707,030 m E

ELEVATION: 75 m DIRECTION: INCLINATION: TOTAL DEPTH: 33 m

HOLE TYPE: Airtrac Percussion DRILLING CONTRACTOR: O'Neill DRILLER:

LOGGED BY: F.T. DATE: 1972 BIT SIZE: TO TO TO

D.G. TONKIN 1979

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:200	SAMPLE NO.	ASSAYED LENGTH	ASSAY VALUE 1978 by A.C.S.H.					
							Cu	Pb	Zn	% Fe	Mn	* Co
	0	RECENT SEDIMENTS 0-1.5 m		0								
		TAPLEY HILL FORMATION 1.5-25.5 m		1.5								
		1.5-4: Magazine Hill clay, khaki		3.0	55334	1.5	40	600	250	1.45	6000	
		4-13: Blue-grey dolomite + minor black, grey & brownish shale. Traces of pyrite & heavy minerals throughout			55335	1.0	20	640	160	1.13	5000	
		Water Table at 5 m.			336		20	160	70	1.74	4500	20
		= laminite facies (dololaminite)			337		40	240	60	1.65	4500	
					338	*	65	400	100	-	-	20
					339		140	120	60	1.65	4500	
					340		210	80	45	1.05	5000	
	10	10-11: Good show of sulphides		10	341		210	80	50	1.05	6500	25
					342	*	450	60	90	-	-	45
					343		520	180	260	1.0	5500	55
		13-25.5: Massive Dolomite. Lt. grey dolomitic limestone to 23m			344	*	92	110	80	-	-	20
					345		30	90	20	0.9	1004	% 15
					346		30	60	35	0.92	9400	
					347		70	100	70	1.12	9000	15
					348		10	60	20	0.84	9750	
					349		10	20	20	0.71	9250	20
					350		30	80	40	0.78	9500	20
	20			20	351		15	80	20	0.7	1004	
					352		25	50	20	0.71	100	%
		:Dolomite pinkish in places			353		30	100	35	0.9	1004	% 15
		From 23m, red quartzite occurring, possibly as pebbles in dolomite.			354		30	100	30	0.82	1118	%
					355		35	50	50	0.9	126	
					356		30	60	45	0.94	122	20
					357		60	70	100	0.72	8000	20
		PANDURRA FORMATION 25.5-33 m			358		40	40	30	0.68	5000	
		Red arenite + variable trace of pyrite.			55359		50	100	65	0.97	6750	15
		All samples contaminated by fallback of dolomite chips			55361	*	110	125	85	-	-	20
	30	Pandurra Fm. Pink quartzite		30	362		30	80	30	0.89	3500	
					363		40	90	70	1.88	4750	10
					364		65	100	80	0.97	4250	
	33	END OF HOLE		33	55365		60	100	80	0.81	3500	10

* 1972 MAGM ASSAY

PACMINEX PTY. LTD.

HOLE No. MONA 9P

PAGE 1 OF 1

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDD: 21/06/72

COMPLETED: 22/06/72

LOCATION: MONALENA Map 6334 L

CO-ORDS: X: 6,487,830 m N Y: 707,795 m E

ELEVATION: 60 m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 15 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: F.T.

DATE: 1972

BIT SIZE:

TO

TO

TO

D.G. TONKIN 1979

[illegible]

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDDER: 22/06/72

COMPLETED: 22/06/72

LOCATION: MONALENA Map 6334 L

CO-ORDS: X 6,488,430 m N Y: 707,535 m E

ELEVATION: 75 m DIRECTION:

INCLINATION:

TOTAL DEPTH: 21 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: F.T. DATE: 1972

BIT SIZE:

TO

TO

TO

D.G. TON/KIN 1979

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 200	SAMPLE No.	Assayed length	ASSAY VALUE					
							Cu	Pb	Zn	% Fe	Mn	* Co
	0	RECENT SEDIMENTS 0-4 m										
		Soil, sand, clay, RQZ & DOL float										
		H.M. tail			55381		60	20	35	0.95	400	
		base sample			382		200	20	40	1.6	200	
		PANDURRA FORMATION 4-21 m			383		20	<20	15	0.65	65	<10
					384		30	60	20	0.9	60	
					385		20	20	10	0.85	60	
		Red - pink			386		10	<20	30	1.05	70	<10
					387		15	<20	20	0.7	50	
	10	4-6: Fractured weathered red quartzite			388		20	<20	30	1.03	45	
					389	*	23	15	20	-	-	<10
					390		20	<20	40	1.16	80	
		6-17: Dark pink quartzite			391		40	<20	35	1.75	55	
					392		-	-	-	-	-	
					393		20	<20	40	1.64	55	<10
					394		15	20	50	1.54	90	
					395		-	-	-	-	-	
		17-21: Pink quartzite			396		20	20	50	1.79	80	<10
					397		20	20	60	1.05	90	
	20				398		30	<20	100	0.97	140	<10
	21	END OF HOLE			55399		30	<20	70	0.65	60	<10

PACMINEX PTY. LTD.

HOLE No. MONA 11P

PAGE 1 OF 1

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDDED: 22/06/72

COMPLETED: 22/06/72

LOCATION: MONALENA Map 6334 G (On road)

CO-ORDS: X: 6,494,620 mNY: 703,970 m E

ELEVATION: 100 m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 27 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: F.T.

DATE: 1972

BIT SIZE:

TO

TO

TO

D.G. JONKIN

1979

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 200	SAMPLE No.	ASSAYED LENGTH	ASSAY VALUE A.C.S.A. 1978					
							Cu	Pb	Zn	% Fe	Mn	* Co
	0	RECENT SEDIMENTS 0-3 m										
		Soil, sand, clay & rubble										
					55400	2.0	20	50	50	1.56	310	
		WHYALLA SANDSTONE 3-10 m			401	1.0	10	20	50	1.4	400	
		3-8: Red brown sandstone. Well rounded grains in red brown matrix.			402		15	20	50	1.94	80	<10
		8-10: Yellow-brown friable lithic slste. Well rounded grains in mustard matrix.			403		<2	20	40	1.64	70	
					404		20	<20	60	0.93	90	
					405		<2	<20	50	1.12	70	
					406		10	<20	50	1.39	80	
	10	TAPLEY HILL FORMATION 10-11.25m			407		10	<20	300	8.35	950	15
		Magazine Hill clay + residual fragments of weathered dolomite			408		20	20	600	21.5	2750	55
		PANDURRA FORMATION 11.25-27 m			409		10	20	150	3.80	320	<10
		Red quartzite			410		10	20	60	2.55	170	<10
		?heavy minerals: 11.25-12m.			411		10	<20	70	2.4	150	<10
		11.25-22m: Oxidized pink quartzite with pinkish-brown matrix cement. Yellow coatings on fracture & grain surfaces.			412		<2	30	60	2.85	110	
					413		5	30	80	2.75	130	<10
					414		10	20	55	2.4	120	<10
					415		10	<20	50	2.9	170	<10
					416		10	20	60	2.9	80	<10
	20				417		20	20	60	1.26	100	
					418		10	<20	40	1.10	60	<10
					419		10	40	50	0.91	30	
		22-27m: Fresh, pink quartzite with typical P.Fm. white clay matrix cement.			420		5	<20	45	0.63	70	
					421		5	40	40	0.39	60	
					422		10	30	45	0.39	20	
					423		10	30	50	0.30	50	
	27	END OF HOLE			55424		<2	<20	40	0.28	35	<10

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDDER: 22/06/72

COMPLETED: 22/06/72

LOCATION: MONALENA Map 6334 G

CO-ORDS: X: 6,494,035 m Y: 704,090 m E

ELEVATION: 100 m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 21 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: F.T. DATE: 1972

BIT SIZE:

TO

TO

TO

D.G. TOWKIN 1979

[illegible]

PROJECT AREA: GUNSON

PROJECT No. 401

SPUDDER: 22/06/72

COMPLETED: 22/06/72

LOCATION: MONALENA Map 6334 G

CO-ORDS: X: 6,493,205 mN^y: 703,980 mE

ELEVATION: 95 m

DIRECTION:

INCLINATION:

TOTAL DEPTH: 21 m

HOLE TYPE: Airtrac Percussion

DRILLING CONTRACTOR: O'Neill

DRILLER:

LOGGED BY: F.T. DATE: 1972

BIT SIZE:

TO

TO

TO

D.G. TONKIN 1979

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 200	SAMPLE No.	Assayed length	ASSAY VALUE A.C.S.L. 1978					
							Cu	Pb	Zn	% Fe	Mn	* Co
	0	RECENT SEDIMENT 0-2 m		0								
		WHYALLA SANDSTONE 2-8 m			55444		22	20	30	0.39	60	
		2-4: Silicified sandstone Silcreted			445		20	20	40	0.32	140	
		4-7: Fine to medium grained S/St.			446		10	20	40	0.25	90	
		Clay content increases with depth			447		22	220	40	0.3	100	
					448		10	220	40	0.28	70	
		TAPLEY HILL FORMATION 8-12 m			449		10	220	40	0.25	85	
		7-13: Magazine Hill clay. Grades			450		10	220	45	0.39	400	<10
	10	from white to greenish & khaki down to grey & black. Talcy feel at top. 10-11m sooty wad. clay dolomitic at base			451		30	120	100	2.4	4.6%	60
					452		40	40	100	5.26	17.2%	130
					453		35	40	80	4.15	4.9%	70
					454		10	20	55	1.8	1.15%	20
		PANDURRA FORMATION 12-21 m			455		10	30	45	0.98	6650	10
		Reddish-brown arenite + differing trace amounts of heavy minerals			456		10	220	40	0.72	2300	
		12-19: Pink quartzite, slightly weathered, with yellowish clay matrix. H.M. 12-14m.			457		10	20	30	0.81	1950	
		19-21: Pink quartzite, fresh, unweathered.			458		10	220	40	0.98	3400	
					459		22	220	40	0.7	2580	<10
					460		30	220	50	0.5	2050	
	20				461		15	220	50	0.72	4200	
	21	END OF HOLE			55462		10	220	40	0.46	2250	<10
							</					

[illegible]

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:200	SAMPLE NO.	ASSAYED LENGTH	ASSAY VALUE A.C.S.A. 1978									
							Cu	Pb	Zn	% Fe	Mn	* Co	* Ag			
	0	RECENT SEDIMENTS 0-1.5 m														
		Sand, soil & rubble (WSs & Silcrete)			55488	1.5	25	20	30	0.6	160					
		TAPLEY HILL FORMATION 1.5-16.5 m														
		1.5-7: Magazine Hill clay. White to light greenish grey; changes to khaki at 6 m. High % of gypsum			489	1.0	5	30	20	0.42	60					
					490		5	20	15	0.51	25					
					491		10	60	20	1.67	600					
					492		20	40	20	1.62	1000					
		7-16.5: Khaki dolomite & clay. Mainly clay, with minor dolomite chips.			493		180	80	100	17.5	1400	25				
		Possibly = Weathered DOLOMITE MEMBER			494	*	155	40	122	-	-	10				
	10				495		130	70	100	16.5	1900	20				
					496		110	20	100	16.5	1550	12				
					497		130	80	90	17.5	1100					
					498	*	3.0	92	38	120	-	-	18			
					55499	1.0	110	40	130	19.0	2600	15				
		Water Table at 16.6 m			55525		60	40	95	1.58	3000	15				
		PANDURRA FORMATION 16.5-30 m			526	*	50	28	68	-	-	10				
		16.5-24: Pale red arenite + HM + limonitic clay filler prob. = oxidized brick red clay.			527		45	40	80	19.0	3550	<10				
	20	Lg brown Fe chips + Lg. RA chips at top suggest broken, oxidized P.F.			528	*	58	48	73	-	-	10				
					529		40	40	80	10.05	2200	10				
		24-30: Deep red arenite + H.M. Hematitic red clay filler			530		20	20	45	6.3	1100	<10				
					531	*	35	53	63	-	-	<10				
					532	*	37	82	55	-	-	<10				
					533		40	20	60	5.55	700	<10				
					534		10	<20	30	3.85	370	<10				
					535		20	<20	60	9.45	1800	<10				
					536		20	40	35	9.4	1650	<10				
					537		10	<20	40	8.55	1650	<10				
	30	END OF HOLE			55538		20	40	40	9.1	1600	<10				

H₂O

MAGAZINE HILL CLAY

PANDURRA FM

* 1972 MGM ASSAY

PACMINEX PTY. LTD.—FIELD DRILL LOG

000057

HOLE No. MORALENA 1

PAGE 1 OF 2

PROJECT: GUNSON 401 DATE STARTED: 18.6.72 TYPE OF DRILL: _____
 AREA: _____ HOLE SIZE: _____
 LOCATION: _____ COORDINATES: _____ CONTRACTOR: _____
 STARTED: _____ ELEVATION: _____ DRILLER: _____
 COMPLETED: _____ DEPTH: 33 m. ReLOGGED BY: D.G. TONKIN (1979)

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS by A.C.S.A.				
					Cu	Pt	Zn	Fe%	Mn
	See earlier log sheets								
	CALNO201C 0-1 m	CZ		1					
	TAPLEY HILL FM. 1-12 m	MAGAZINE HILL clay	69905	1	-	-	-	-	-
3			906	1	-	-	-	-	-
	Gypsum + white clay + many yellow dol. chips + sand. Brown + khaki dolomite chips		907	1	25	80	50	0.60%	3000
			908	1	90 85	70 45	50 40	1.20	6250
6	Brown + yellow brn. dol. chips		69909	1	90	70	70	2.55	8000
	Lt. brown dol. Sand Mn chips		910	1	110 63	90 40	70 35	2.45	1.10%
	As above		911	1	45	100	40	2.75	1.13%
9			912	1	-	-	-	-	-
	Khaki-brn. dol. finely pulverised Mn		913	1	35 42	50 32	50 50	8.10	1.26%
	Orange Khaki dol. large chips Mn		914	1	60 60	100 30	80 45	26.5	3000
12	Khaki dol. + white to salmon oxid. az + clay.		915	1	40	70	80	13.5	2000
	PANDURRA FM. 12-33 m		916	1	40 65	60 35	80 63	13.0	3000
	Pale az + yellow ochre clay + black H.M. tail		917	1	60 43	40 40	75 60	11.0	3000
	As above. Prob. fractured. + leached az		69918	1	80 38	40 30	70 50	8.10	2750
15	As above.		919	1	50 42	60 40	75 65	8.00	3500
	As above		920	1	40 55	40 25	40 60	6.65	2250
	As above. Clay has become gradually less yellow. H.M. tail has slowly declined		921	1	40 82	20 30	60 55	7.20	2750
18	As above		922	1	125 48	70 40	70 55	6.70	2500
	As above (+ minor dol. content)		923	1	95 38	40 25	60 35	5.20	1650
21	White az + yellow-brown clay + some H.M.		924	1	70 28	20 20	100 28	3.90	2250
	As above		925	1	32	20	40		
	As above		926	1	30 52	20 25	40 42	5.35	2500
24	As above		927	1	90 40	30 20	60 80	5.00	3000
	As above		69928	1	30 30	40 35	45 40	4.50	1450
	As above		929	1	40 35	20 25	40 31	4.30	1600
27	As above - clay changing to pinkish brown. white az apparently fractured.		930	1	30 31	50 50	35	4.85	3000
	As above		931	1	90 23	50 15	50 27	3.80	1300
	As above		932	1	120 30	40 20	65 30	3.50	1900
30	As above		69933	1	35 25	40 20	40 32	3.10	2750

N.B. lower figures are previous MGM assays.

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PAGE 2 OF 2

[illegible]

PROJECT:.....Gunson 401..... DATE STARTED:18. 6. 72..... TYPE OF DRILL:

AREA:..... COORDINATES: HOLE SIZE:

LOCATION:..... CONTRACTOR:

STARTED:..... ELEVATION: DRILLER:

COMPLETED:..... DEPTH:21 m..... RELOGGED BY: D.G. TONKIN 1979

[illegible]

PACMINEX PTY. LTD.—FIELD DRILL LOG

000060

HOLE No. MONGLENA 3
PAGE 1 OF 1

PROJECT: GUNSON 401 DATE STARTED: 18.6.72 TYPE OF DRILL: _____
AREA: _____ HOLE SIZE: _____
LOCATION: _____ COORDINATES: _____ CONTRACTOR: _____
STARTED: _____ ELEVATION: _____ DRILLER: _____
COMPLETED: _____ DEPTH: 27 m RELOGGED BY: D.R. IDYKIN 1979

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS				
					Cu	Pb	Zn	Fe%	Mn
	CARNOTZIC 0 - 1.5 m	Recent		1					
	Red clay, RQ + black shale chips. Minor dol. Abund. HM								
	TAPLEY HILL FM. 1.5 - 3 m		69957	1	90	40	60	6.55	1.22%
3	Shale or laminite facies		958	1	-	-	-	-	-
	PANDURRA FM. 3 - 27 m.		959	1	-	-	-	-	-
	Pink RQ chips + abund. H.M. tail in brown matrix		960	1	70	40	40	4.40	900
6	Fractured RQ + yellowish clay		961	1	25	30	10	2.50	400
	As above		962	1	30	15	25		
	RQ + red/brown clay		963	1	70	40	80	3.45	230
9	Pulv. RQ + red/brown clay		964	1	25	60	20	5.80	180
	Fract. + pulv. RQ + r/b clay		965	1	30	40	10	3.30	300
			966	1	31	50	30		
			967	1	130	80	60	3.00	500
12	RQ + dk. pink clay Pulverised		968	1	-	-	-	-	-
	As above		969	1	30	80	25	2.50	550
	RQ + dk. pink clay - micaceous lustre.		970	1	55	80	30	2.10	310
15	As above dk. pink or brick red? Pulv.		971	1	29	70	35		
	As above		972	1	20	90	10	2.20	330
			973	1	20	65	15		
18	As above		974	1	65	100	30	3.10	300
			975	1	19	70	20		
			976	1	15	120	30	1.85	110
			977	1	30	90	30		
			978	1	22	65	23		
21	As above		979	1	30	80	10	2.65	110
	As above		980	1	25	45	20		
			981	1	23	70	25		
			982	1	15	60	10	2.30	50
24	As above		983	1	28	55	15		
	As above		984	1	20	60	20	2.20	140
			985	1	15	160	30	2.65	70
			986	1	35	140	21		
			987	1	20	280	85	2.30	130
27	As above		988	1	15	120	25	1.60	65
	Darken.		989	1	25	130	41		
	V. dark purple - red colour. Finally pulv.		990	1	10	90	20	3.55	90
	As above		991	1	28	70	22		
			992	1	15	60	25	6.10	300
			993	1	13	35	33		
	Back to dk. pink - brick red.		994	1	30	440	30	3.15	280
	END HOLE		995	1	42	390	15		
					705		600		
					21% = 0.8				

PACMINEX PTY. LTD.—FIELD DRILL LOG

000061

PROJECT: GUNSON 401 DATE STARTED: 19. 6. 72 TYPE OF DRILL: _____
AREA: _____ HOLE SIZE: _____
LOCATION: _____ COORDINATES: _____ CONTRACTOR: _____
STARTED: _____ ELEVATION: _____ DRILLER: _____
COMPLETED: _____ DEPTH: 45 m RE LOGGED BY: D.G. TONKIN 1979

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS					
					Cu	Pb	Zn	Fe%	Mn	Co
3	CARNOZONIC 0 - 2 m	Recent		1						
				1						
	TAPLEY HILL FM. 2 - 45 m									
6	Sandy white clay	W. ss.	69983	1	15	30	20	0.65	240	
	gls. gr. lam. wss + silic. wss. chips + gypsum + red clay		984	1	20	60	30	0.90	400	
	wss gr. + khalsi weathered shale.	Oxide	985	1	20	180	170	2.60	1650	
9	Khalsi dol. + shale (+ clay?) (986)		986	1	55	195	180			20
		FORMATION LAMINITE			20	380	250	2.80	2750	
			987	1	41	300	470			25
12	Dk. grey shale + lt. grey dolomite = laminite	HILL	988	1	40	220	750	2.75	1900	
	As above.		989	1	45	150	660			15
		TAPLEY	69990	1	70	170	840			20
15			991	1	100	160	450	3.05	950	
		HILL	992	1	50	130	580			20
	Dk. grey shale >> lt. grey dolomite.		993	1	60	160	220	3.50	1000	
18	As above	TAPLEY	994	1	74	135	200			20
	As above		995	1	50	160	210	3.50	1100	
		HILL	996	1	55	135	220			25
21			997	1	50	120	300	3.25	1150	
		TAPLEY	998	1	52	135	275			20
	Dk. grey shales ± v.v. minor dolom.		999	1	-	-	-	-	-	
24		HILL	995	1	60	130	265			23
			996	1	30	140	360	3.35	1150	
		TAPLEY	997	1	58	140	325			25
27			998	1	49	125	340			20
		HILL	999	1	60	160	380	3.35	1800	
	Dk. grey shale ± v. minor dolomite. Minor sand grs. (contamin?)		55201	1	65	125	400			20
30	As above	TAPLEY	55202	1	40	120	270	2.95	1750	
	As above		55203	1	60	115	315			22
		HILL	55204	1	-	-	-	-	-	
33			55205	1	110	100	335			20
		TAPLEY	55206	1	-	-	-	-	-	
	Dk. grey shale ± v. minor dolomite. Minor sand grs. (contamin?)		55207	1	35	90	310			18
36	As above	HILL	55208	1	43	130	350			20
	As above		55209	1	55	140	400	2.80	1450	
		TAPLEY	55210	1	60	130	460			21
39			55211	1	45	120	350	2.85	1800	
		HILL	55212	1	38	110	315			20
			55213	1	35	80	260	2.90	1650	
42		TAPLEY	55214	1	52	85	250			25
			55215	1	35	90	170	2.80	1100	
		HILL	55216	1	42	110	230			20
45			55217	1	40	100	290	2.40	1100	
		TAPLEY	55218	1	49	85	200			20
	As above. Contamination ~ 15%		55219	1	35	80	190	2.50	1000	
48		HILL	55220	1	35	70	295			20
			55221	1	60	160	600	3.75	1250	
		TAPLEY	55222	1	50	70	210			20
51			55223	1	40	60	220	2.75	950	
		HILL	55224	1	32	65	220			15
			55225	1	32	65	220			15

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[illegible]

PACMINEX PTY. LTD.—FIELD DRILL LOG

000063

HOLE No. Monalewa S
PAGE 1 OF 2

PROJECT: GUNSON 401 DATE STARTED: 19.6.72 TYPE OF DRILL:
AREA: HOLE SIZE:
LOCATION: COORDINATES: CONTRACTOR:
STARTED: ELEVATION: DRILLER:
COMPLETED: DEPTH: 45 m. RE LOGGED BY: D.G. TONKIN 1979

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS					
					Cu	Pb	Zn	Fe%	Mn	Co
3	WHYAKKA SANDSTONE - 9.5	Dune sand		1						
	Orange dune sand.		SS227	1	15	20	20	0.97	120	
	As above.		228	1	40	20	20	0.40	45	
6	Buff coloured weathered s'st., prob. silicified	WHYAKKA SS	229	1	15	20	20	0.24	45	
	Pale orange-pink silicified lithic s'st.		230	1	10	<20	10	0.23	20	
	As above. Clayey, gypsiferous.		231	1	15	<20	20	0.19	15	
9	Fine - Med. gr. lithic ss to v. well round gr. & white matrix friable.	WHYAKKA SS	232	1	10	<20	10	0.12	20	
	As above, finer grained		233	1	10	40	10	0.17	20	
	Fine gr. s'st to v. abundant white clay.		234	1	10	<20	10	0.20	50	
12	TARKEY HILL FM. 9.5 - 45 m.	WHYAKKA SS	235	1	10	30	35	0.18	20	
	As above		236	1	15	30	10	0.17	20	
	White clay + sand grains.		237	1	10	<20	30	0.17	20	
15	As above	MAGAZINE HILL CLAY	238	1	30	20	20	0.18	50	
	As above.		239	1	45	40	10	0.37	20	
	Pale yellow white clay		SS240	1	25	120	20	0.43	15	
18	Pale grey white clay	MAGAZINE HILL CLAY	241	1	25	120	20	1.36	20	
	As above.		242	1	70	90	20	1.05	40	
	Pale grey - white clay		243	1	115	80	30	1.36	120	
21	As above	MAGAZINE HILL CLAY	244	1	310	120	180	1.56	150	
	Khaki fawn clay		245	1	90	240	500	1.76	1050	
	Fawn - khaki clay		246	1	-	-	-	-	-	
24		MAGAZINE HILL CLAY	247	1	120	900	500	1.60	3500	
			248	1	130	600	500	1.60	2750	
	Fawn clay - Khaki		249	1	100	160	240	1.50	1600	
27	Khaki clay & weathered shale.	THE SHALE	SS250	1	40	180	360	1.90	2500	
	Dark grey laminated shale.		251	1	30	220	390	1.70	1300	
	As above		252	1	49	200	250			25
30	As above	THE SHALE	253	1	40	160	210	1.56	1650	
	As above		254	1	52	200	245			25
	As above		255	1	45	180	200	1.66	1700	
30	As above	THE SHALE	256	1	60	170	420			25
	As above		257	1	45	180	200	1.66	1700	
	As above		258	1	40	140	270	1.50	1600	
30	As above	THE SHALE	259	1	43	210	365			20

PROJECT:.....

DATE STARTED:

TYPE OF DRILL:

AREA:.....

COORDINATES:

HOLE SIZE:

LOCATION:.....

ELEVATION:.....

CONTRACTOR:

STARTED:.....

DEPTH: 45 m.

DRILLER:

COMPLETED:.....

LOGGED BY:

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS					
					Cu	Pb	Zn	Fe %	Mn	Co
33	TAPLEY HILL Fm. (cont.) to 45m	T.H.F. SHALE	SS256	1	35	190	240	1.46	1700	
			45	170	275			20		
	DK. grey shale		257	1	35	140	240	1.60	2000	
	As above			41	220	430		15		
36	As above		258	1	55	120	300	1.46	1850	
	DK. grey shale.		259	1	45	190	500	1.40	1750	
			35	150	400		20			
	DK. grey shale + grey dolom. = laminite ?		260	1	60	120	400	1.46	3000	
39		TAPLEY HILL LAMINITE			45	220	600		20	
				115	160	310	1.30	1600		
	DK. grey shale + grey dolom. = laminite			261	1					
	As above			262	1	40	180	390	1.33	2000
42					40	140	700		20	
	DK. grey shale + minor grey dolomite		263	1	30	120	500	1.26	2000	
	DK. grey shale.		264	1						
			40	140	550	1.28	1900			
45					42	140	720		20	
	DK. grey shale + v. minor grey dol.		265	1	60	160	340	1.42	1800	
	DK. grey shale + minor grey dolom.		266	1	45	260	400	1.40	1600	
	DK. grey shale + v. minor dol.		50	300	560		25			
45					35	180	500	1.36	1650	
	As above		267	1						
			268	1	50	160	400	1.42	1400	
			45	215	430		23			
45					35	160	260	1.26	1500	
	As above.		SS270	1	40	120	290	1.33	1400	
					52	215	325		25	
	END HOLE									

PROJECT: GUNSON 401 DATE STARTED: 20.6.72 TYPE OF DRILL:
AREA: HOLE SIZE:
LOCATION: COORDINATES: CONTRACTOR:
STARTED: ELEVATION: DRILLER:
COMPLETED: DEPTH: 36 m. RELOGGED BY: D.G.T. (1979)

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS					
					Cu	Pb	Zn	Fe%	Mn	Co
	CAINOZOIC 0 - 2 m	Recent		1						
	TAPLEY HILL FM. 2 - 27 m.			1						
3	Sand, dol. chips = khaki - brown clay.	MAGAZINE HILL clay	55271	1	70	60	240	1.78	4500	
	2 to 5.5: Magazine Hill Clay		272	1	50	100	230	1.40	7500	
	Fawn clay - oxidized shale.		273	1	140	120	210	1.50	5750	
6	As above.		274	1	400	90	190	1.55	4750	
	5.5 to 27: Woodculla Dolomite Member		275	1	-	-	-	-	-	85
	As above + chips yellow - brn. dolomite		276	1	190	40	170	1.15	8500	
9	Reddish = brown dolom. ± dendritic Mn		277	1	180	80	230	0.96	7250	
	As above.		278	1	90	30	100	0.79	7000	
	Pink dolomite ± dendritic Mn.		279	1	-	-	-	-	-	25
12	Tr. v. lt. grey shale		55280	1	130	45	115			
	Pink, brown + v. light grey dolom. + Mn n	FORMATION	281	1	80	20	80	0.77	7750	
	Fawn, pink + lt. grey dolom. ± Mn. Tr. grey shale		282	1	50	50	60	0.83	8750	
15	DK., khaki Mn dolom. Tr. pink dolom.		283	1	45	40	55	0.71	8250	
	DK., khaki, Mn dolom.		284	1	80	30	85	0.76	8500	
	Pink, fawn = light grey Mn. dolom.		285	1	-	-	-	-	-	15
18	Buff Mn dolom. + minor lt. grey dolomitized shale		286	1	65	20	45	0.72	8500	
	Pink Mn dolom.		287	1	60	40	40	0.63	6750	
	Fairly fresh, buff to grey dolomite		288	1	70	70	55	0.79	0.94	
21	Fresh lt. grey dolomite		289	1	-	-	-	-	-	20
	Fresh lt. & mid grey dolomite		55290	1	45	55	45			
24	As above.		291	1	30	80	35	0.68	1.08	
		TAPLEY	292	1	30	80	40	0.77	1.04	
			293	1	10	20	30	0.68	1.22	
27			294	1	-	-	-	-	-	25
			295	1	40	55	40			
			296	1	30	60	45	0.92	1.14	
30			297	1	20	20	35	0.76	1.32	
			298	1	-	-	-	-	-	30
			299	1	37	60	40			
36	PANDURRA FM. 27 - 36 m.		55298	1	28	40	35			10
		P. FM.	299	1	12	10	12			<10
			300	1	-	-	-	-	-	<10

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CSR/0035

PACMINEX PTY. LTD.—FIELD DRILL LOG

000067

HOLE No. **MONALENA 7**
PAGE **1** OF **1**

PROJECT: GUNSON 1601 DATE STARTED: 20.6.72 TYPE OF DRILL:
AREA: HOLE SIZE:
LOCATION: COORDINATES: CONTRACTOR:
STARTED: ELEVATION: DRILLER:
COMPLETED: DEPTH: 30 m RELOGGED BY: D. G. JONKIN (1979)

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978	ASSAYS by A.C.F.L.	Cu	Pb	Zn	Fe %	Mn	Co
	CAINOZOIC 0 - 2.5 m			1								
	TAPLEY HILL Fm. 2.5 - 10.5 m	Recent		1								
	Wocalla Dolomite Member											
3	Khaki, Mn-speckled oxid. dolom. + red clay G ₂		SS305	1	20	40	30	1.34	1.30			
	Khaki, Mn-speckled oxid. dolom.		306	1	20	60	35	1.13	1.42			
			307	1	20	45	25					20
6	As above + minor white clay.		308	1	30	20	30	0.77	1.46			
	As above.		309	1	20	30	40	0.82	1.30			
	DK. khaki-green - black Mn dolom. + recent clay cont.		SS310	1	45	40	30	0.80	1.32			25
	DK. green-black Mn dolom. + minor white clay. (311)				29	55	30					
9	DK. green-black Mn dolom. (312)		311	1	45	50	30	1.03	1.46			
	PANDURRA Fm. 10.5 - 30 m.		312	1	30	30	30	0.85	1.32			
	Mn dolom. as above + red quartzite chips PF.		313	1	30	40	30	0.75	8500			10
	DK. pink Q ₂ + white + pale green soft clay				32	40	25					
12	'Liller' (small chips or flakes 1-2 mm.)		314	1	15	20	20	0.46	1550			<10
					38	20	15					
	DK. pink PF Q ₂ (NB white kaolin matrix cement)		315	1	32	20	25					<10
			316	1	10	20	20	0.54	600			
15	As above		317	1	10	30	20	0.69	50			<10
	As above				35	20	15					
	As above		318	1	20	60	20	0.69	450			<10
					20	22	20					
18	Brick red Q ₂ Micaceous lustre (NB white kaolin cement)		319	1	10	20	10	0.70	450			<10
	As above				25	30	20					
	As above		SS320	1	25	20	20	1.54	600			<10
					35	15	15					
			321	1	10	20	10	1.28	300			<10
					25	20	20					
21			322	1	10	30	10	1.34	450			<10
					27	<10	15					
			323	1	53	10	40					15
	Pink quartzite PF + white kaolin cement.				10	40	30	0.97	350			<10
	As above		324	1	21	15	20					
24	As above				20	40	20	1.03	200			<10
	As above		325	1	35	10	20					
	As above				10	20	20	1.11	300			<10
	Dark pink Q ₂ .		326	1	32	10	15					
27	As above				10	20	20	1.17	400			<10
			327	1	61	15	40					
			328	1	5	20	15	1.59	200			<10
					19	10	10					
			329	1	10	20	15	1.42	250			<10
					19	15	<10					
	Pink Q ₂		SS331	1	5	30	20	1.13	150			<10
					20	20	10					
	Pink Q ₂		332	1	20	20	20	1.09	350			<10
					21	20	10					
30	Dark red - red brown Q ₂ . 4M. tail = drill steel		SS 333	1	10	40	10	1.78	300			10
					21	15	10					

END OF HOLE

200

2 = 1.43

PACMINEX PTY. LTD.—FIELD DRILL LOG

PROJECT: GUNSON 401 DATE STARTED: 20. 6. 72 TYPE OF DRILL:
AREA: HOLE SIZE:
LOCATION: COORDINATES: CONTRACTOR:
STARTED: ELEVATION: DRILLER:
COMPLETED: DEPTH: 33 m RELOGGED BY: D.G. JOWKIN (1979)

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS					
					Cu	Pb	Zn	Fe%	Mn	Co
0	CAINOZOIC 0 - 1.5 m	Recent								
3	TAPLEY HILL FM. 1.5 - 25.5m Khaki shaley clay Grey clay + khaki + brown dolomite chips lt. grey + fawn dol. chips + some dk. grey clay	MARGALINE HILL CLAY 4.0	SS334	1.5	40	600	250	1.45	6000	
			335	1	20	640	160	1.13	3000	
6	Grey dolomite chips + some dark grey partings. Tabular		336	1	20	160	70	1.74	4500	
			337	1	30	160	70			20
			338	1	40	240	60	1.65	4500	
			339	1	-	-	-	-	-	
					65	400	100			20
			340	1	140	120	60	1.65	4500	
9	lt. grey massive dol. chips + dk. grey shale chips i.e. laminite facies lt. grey dol. chips + dk. grey shale chips Mainly lt. grey dol. + little grey shale.		SS340	1	210	80	45	1.05	5000	
			341	1	210	80	50	1.05	6500	
					138	65	50			25
			342	1	-	-	-	-	-	
					450	60	90			45
12	lt. grey dol. + some dk. grey partings tabular		343	1	520	180	260	1.00	5500	
					620	125	290			55
			344	1	-	-	-	-	-	
					92	110	80			20
			345	1	30	90	20	0.90	1104	
					60	45	50			15
15	Massive clean, lt. grey dolomite brittle chips 'limestone' v. angular chips As above As above As above		346	1	30	60	35	0.92	0.94	
			347	1	70	100	70	1.12	9000	
					42	90	70			15
			348	1	10	60	20	0.84	9750	
18	As above As above As above		349	1	10	20	20	0.71	9250	
					22	45	20			20
			SS350	1	30	80	40	0.78	9500	
					35	65	35			20
			351	1	15	80	20	0.70	1.04	
21	As above As above As above		352	1	25	50	20	0.71	1.00	
			353	1	30	100	35	0.90	1.04	
					50	55	45			15
			354	1	30	100	30	0.82	1.18	
24	Nearly white massive brittle dolomitic limest. As above As above 50% + PFm. Red quartzite. Malachite seen assoc. + RA in sample		355	1	35	50	50	0.90	1.26	
			356	1	30	60	45	0.94	1.22	
					39	40	50			20
			357	1	60	70	100	0.72	8000	
					85	65	50			20
27	PANDURRA FM. 25.5 - 33 m Pink Q2 + ~30% pale grey dol. frag. contain. Pink Q2 + 30% contain. + lt. grey dol., dk. grey dol., brown dol. + grey shale. Red Q2 + ~20% dolom. + grey shale contain.	PANDURRA FM. QUARTZITE 25.5	358	1	40	40	30	0.68	5000	
			359	1	50	100	65	0.97	6750	
					50	80	55			15
			SS361	1	-	-	-	-	-	
					110	125	35			20
30			362	1	30	80	30	0.89	3500	

PAGE 2 OF 2

CSR/0035

000070

PAGE 1 OF 1

TYPE OF DRILL:

HOLE SIZE:

CONTRACTOR:

DRILLER:

RECORDED BY: D. G. TONKIN (1978)

DEPTH: 15 m RELOGGED BY: D.G. TONKIN (1978)

[illegible]

PACMINEX PTY. LTD.—FIELD DRILL LOG

000071

PROJECT: GUNSON 401 DATE STARTED: 22. 6. 72 TYPE OF DRILL:
AREA: COORDINATES: HOLE SIZE:
LOCATION: CONTRACTOR:
STARTED: ELEVATION: DRILLER:
COMPLETED: DEPTH: 21 m. RE LOGGED BY: D.G. JONKIN (1979)

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS						
					Cu	Pb	Zn	Fe %	Mn	Co	
3	CALCINOSITE 0 - m	Recent		1							
			1								
	Red/brown sandy clay		55381	1	60	20	35	0.95	400		
6	Qzite, gypsum, Khatci dol. (minor chips) + H.M.		382	1	200	20	40	1.60	200		
	PANDURRA Fm. 4 - 21 m		383	1	20 40	<20 15	15 20	0.65	65	<10	
	Fractured, oxidized RQ chips + gypsum (383)		384	1	30	60	20	0.90	60		
9	Fractured, weakened RQ chips	QUARTZITE	385	1	20	20	10	0.85	60		
	Pulverised pink quartzite. Some chips		386	1	10 28	<20 10	30 15	1.05	70	<10	
	Pulverised dark pink quartzite		387	1	15	<20	20	0.70	50		
12	As above	Fm.	388	1	20	<20	30	1.03	45		
	As above		389	1	23	15	20	-	-	<10	
	As above		55390	1	20	<20	40	1.16	80		
15	As above .	PANDURRA	391	1	40	<20	35	1.75	55		
	As above . Prob. red & white banded.		392	1	-	-	-	-	-		
	As above		393	1	20 24	<20 15	40 15	1.64	55	<10	
18	As above		394	1	15	20	50	1.54	90		
	As above		395	1	-	-	-	-	-		
	Pulverised pink quartzite		396	1	20 25	20 10	50 15	1.79	80	<10	
21	As above		397	1	20	20	60	1.05	90		
	As above		398	1	30 25	<20 20	100 15	0.97	140	<10	
	As above		55399	1	30 30	<20 10	70 15	0.65	60	<10	
	END OF HOLE										

PACMINEX PTY. LTD.—FIELD DRILL LOG

HOLE No. **MONALENA 11**
PAGE **1** OF **1**

000072

PROJECT: GUNSON

DATE STARTED: 22.6.72

TYPE OF DRILL: _____

AREA: _____

COORDINATES: _____

HOLE SIZE: _____

LOCATION: _____

CONTRACTOR: _____

STARTED: _____

ELEVATION: _____

DRILLER: _____

COMPLETED: _____

DEPTH: 27 m

RELOGGED BY: D.G. JONKIN (479)

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS					
					Cu	Pb	Zn	Fe %	Mn	Co
3	CAINOZOIC 0 - 3 m	Recent		1						
				↑						
				1						
			SS400	2	20	50	30	1.56	310	
6	WHYALLA SANDSTONE 3 - 10 m	Sandstone	401	1	10	20	50	1.40	400	
	Well rounded grains in red/bm. clay matrix		402	1	15	<20	50	1.94	90	<10
	As above		403	1	<2	<20	40	1.64	70	
	As above		404	1	20	<20	60	0.93	90	
9	As above	Whyalla	405	1	<2	<20	50	1.12	70	
	As above		406	1	10	<20	50	1.39	80	
	Frangible, well rounded grs. in yellow/brown matrix.		407	1	10	<20	300	8.35	950	15
	Frangible, well rounded grs. in mustard matrix				20	15	80			
12	TAPLEY HILL Fm. 10 - 11.25 m	MAGAZINE HILL CLAY	408	1	20	20	600	21.5	2750	55
	Dk. Khaki/bm. Mn Clay.				24	20	220			
	Tr. white dol. + white clay. Mostly pink quartzite. Mantle		409	1	10	20	150	3.80	320	<10
					19	15	30			
15	PANDURRA Fm. 11.25 - 27 m.	FORMATION QUARTZITE.	SS410	1	10	20	60	2.55	170	<10
	Oxid. pink qz. ± yellowish-brown matrix cement				21	15	20			
	As above		411	1	10	<20	70	2.40	150	<10
	As above				22	<10	45			
18	As above	PANDURRA	412	1	<2	30	60	2.85	110	
	As above		413	1	5	30	80	2.75	130	<10
	Oxid. red qz. ± pinkish-brown matrix cement.				23	<10	20			
	Some yellow chip coatings.		414	1	10	20	55	2.40	120	<10
21	Oxid. Rq ± orange-brown matrix cement	PANDURRA			20	10	10			
	As above		415	1	10	<20	50	2.90	170	<10
	As above				25	12	20			
	As above		416	1	10	20	60	2.90	80	<10
24	As above	PANDURRA			20	15	20			
	As above		417	1	20	20	60	1.26	100	
	As above		418	1	10	<20	40	1.10	60	<10
	As above, becoming fresher				30	12	25			
27	Oxid. Rq ± pinkish matrix cement. Some yellow stainings.	PANDURRA	419	1	10	40	50	0.91	30	
	Fresh, pink quartzite ± typical P.Fm. white matrix cement.		SS420	1	5	<20	45	0.63	70	
	As above				5	40	40	0.39	60	
	As above		421	1	10	30	45	0.39	20	
	As above				10	30	50	0.30	50	
	As above		422	1						
	As above		423	1						
	As above		SS424	1	<2	<20	40	0.28	35	<10
	As above				20	30	15			
	END OF HOLE									

000073

PAGE 1 OF 1

CSR/0035

PAGE 1 OF 1

[illegible]

PACMINEX PTY. LTD.—FIELD DRILL LOG

HOLE No. MONALENA 14

000075

PAGE 1 OF 1

PROJECT: GUNSON HOLE

DATE STARTED: 23 - 6 - 72

TYPE OF DRILL:

AREA:

COORDINATES:

HOLE SIZE:

LOCATION:

CONTRACTOR:

STARTED:

ELEVATION:

DRILLER:

COMPLETED:

DEPTH: 27m

RELOGGED BY: D.G. JONKIN (1979)

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1979 ASSAYS					
					Cu	Pb	Zn	Fe %	Mn %	Co
	CAINOZOIC 0 - 2	Recent		1						
				1						
	Brown Mn Dol. Green-grey dust.	WOODCOTT DOLOMITE MEMBER	55463	1	20 49	20 38	60 50	0.93	1.26	30
3	TAPLEY HILL FM. 2 - 18.5		464	1	30 55	40 45	65 40	1.11	1.88	25
	WOODCOTT Dolomite Member		465	1	25	40	55	1.02	1.80	
	As above		466	1	30 58	50 45	55 35	1.05	1.72	15
6	Yellow-brown Mn Dol. Lt. brown dust. (oxid.)		467	1	5	40	55	1.17	1.82	
	As above		468	1	30	30	50	0.88	1.33	
	As above		469	1	10 40	40 40	60 30	0.81	1.38	20
9	As above		55470	1	10	20	50	0.79	1.37	
	As above		471	1	10	20	50	1.01	1.28	
12	As above		472	1	10	60	45	0.98	1.18	
	Pinkish/brown Mn Dol. Fawn dust. (oxidised.)	WOODCOTT DOLOMITE MEMBER	473	1	10	20	50	1.35	1.47	
	Y. fresh fawn/wht. - grey/wht. Dol. Pale fawn (greyish) dust.		474	1	10 33	40 40	60 20	0.86	1.30	20
15	Y. fresh grey/white Lt. Dol. Off-white dust.		475	1	10	40	50	1.16	1.32	
	Buff-brown Mn Dol. Khaki dust. (oxidised)		476	1	10 38	40 40	65 25	1.96	2.40	15
	As above		477	1	15 42	100 50	55 35	1.88	2.90	20
18	Pinkish + black Mn. Dol. Khaki dust. (oxid)		478	1	10 50	50 55	120 115	4.20	8.10	30
			479	1	-	-	-	-	-	25
	PANDURRA FM. 18.5 - 27	PANDURRA FORMATION	55480	1	30 43	60 45	150 100	5.55	4.90	20
21	Pink Qz + >50% Mn Dol. + Khaki dust contam.		481	1	15 40	30 25	70 65	2.85	1.88	15
	Poly. Qz + 10% black gr. contam. + green Khaki mud (contam?)		482	1	30 40	40 35	140 110	5.90	3.3	20
	Poly. Qz gr. + d. brn. mud + 10% black chip contam?		483	1	10 44	20 25	70 70	3.25	1.18	15
24	Pink Qz chips/gr. + brn. mud + 2% dol. chip cont?		484	1	10 37	20 15	50 45	2.30	1.38	10
	Oxid. Qz grains + brown mud matrix + 10% Dol. contam.		485	1	30 50	20 20	75 45	2.50	1.79	10
	Oxid. Qz grains + d. brown muddy matrix + minor dol. chip contam.		486	1	20 42	20 20	60 45	2.35	1.25	10
27	Red Qz gr. + abund. dk. brown muddy matrix		55487	1	20 50	20 15	60 45	3.65	1.18	10
	END OF HOLE									

PACMINEX PTY. LTD.—FIELD DRILL LOG

000078

PROJECT: GUNSON H01 DATE STARTED: 23. 6. 72 TYPE OF DRILL:
AREA: COORDINATES: HOLE SIZE:
LOCATION: CONTRACTOR:
STARTED: ELEVATION: DRILLER:
COMPLETED: DEPTH: 30 m RELOGGED BY: D.G. TARKIN (1979)

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS					
					Cu	Pb	Zn	Fe%	Mn	Co
	CAINUZOIC 0 - 1.5	Recent	N.S.	1.5						
3	TAPLEY HILL FM. 1.5 - 16.5	MAGAZINE HILL CLAY	55488	1.5	25	420	30	0.60	160	
	Gypsum + white & pink clay		489	1	5	30	20	0.42	60	
	Gypsum + pink & white clay		490	1	5	420	15	0.51	25	
	Gypsum + white kaolin clay		491	1	10	60	20	1.67	600	
6	White kaolin clay + some gypsum		492	1	20	40	20	1.62	1000	
	White & pale brown clay + gypsum.		493	1	180	80	100	17.5	1400	
	Yellow ochre - khaki + white clay				220	50	100			25
9	Yellow ochre clay		494	1	155	40	122			10
	Yellow ochre clay		495	1	130	70	100	16.5	1900	
	Khaki brown clay				120	45	88			20
			496	1	110	20	100	16.5	1550	
					110	45	80			12
12	Khaki brown clay (weathered shale)		497	1	130	80	90	17.5	1100	
			498	3	92	38	120			18
15	Khaki brown clay + dolom. + Fe	FORMATION	55499	1	110	40	130	19.0	2600	
	Khaki - yellow ochre clay + Fe + RA + dol.				105	60	118			15
			55525	1	60	40	95	1.58	3000	
					75	40	90			15
	PANDURRA FM. 16.5 - 30		526	1	50	28	68			10
	Oxid. RA chips + Fe chips + yellow clay		527	1	45	40	80	19.0	3550	
					50	30	60			10
	55528 missing		528	1	58	48	73			10
					40	40	80	10.05	2200	
			529	1	40	51	92			10
21	Pink granite + Fe + yellow clay filler + h.m.				20	20	45	6.30	1100	
	Pink granite + Fe + yellow clay + h.m.		530	1	32	48	65			10
					32	48	65			10
	55531 missing		531	1	35	53	63			10
					35	53	63			10
24	55532 missing	Fe PANDURRA	532	1	37	82	55			10
	Brick red az + h.m.				40	420	60	5.55	700	
			533	1	25	35	45			10
					10	420	30	3.85	370	
	Brick red az + h.m.		534	1	28	23	40			10
					20	420	60	9.45	1800	
27	Brick red az. + h.m.		535	1	34	35	85			10
					20	40	35	9.40	1650	
	Brick Red az + h.m.		536	1	30	33	62			10
					10	420	40	3.55	1650	
	Brick red granite + h.m.		537	1	29	30	55			10
30					20	40	40	9.10	1600	
	Homotized Brick Red az ± abund. H.M. tail		55538	1	32	32	55			10

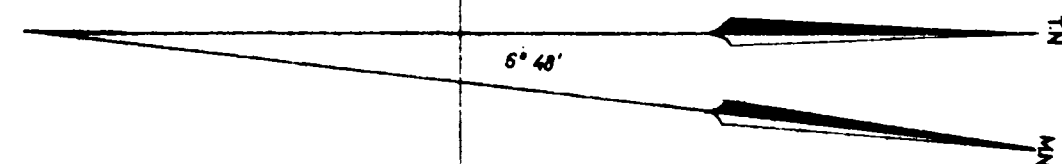
END OF HOLE

PACMINEX PTY. LTD.—FIELD DRILL LOG

000077

PROJECT: GUNSON 401 DATE STARTED: 24. 6. 72 TYPE OF DRILL: _____
AREA: _____ COORDINATES: _____ HOLE SIZE: _____
LOCATION: _____ CONTRACTOR: _____
STARTED: _____ ELEVATION: _____ DRILLER: _____
COMPLETED: _____ DEPTH: 19 m LOGGED BY: _____

Metres	DESCRIPTION		SAMPLE No.	Assay Length	1978 ASSAYS					
					Cu	Pb	Zn	Fe%	Mn	
3	CAINOZOIC 0-19	Sand + Pecc 10-15		1.5						
			SS539	1.5	15	<20	30	0.82	95	
			S40	1.5	20	<20	20	0.84	120	
6			S41	1.5	10	<20	20	1.32	170	
			S42	1.5	10	30	30	1.38	75	
			S43	1.5	10	<20	20	0.66	50	
9			S44	1.5	5	20	30	0.88	50	
			S45	1.5	10	<20	20	0.82	40	
			S46	1.5	10	20	30	1.02	70	
12			S47	1.5	10	<20	15	0.95	80	
			S48	1.5	15	20	10	1.16	190	
			S49	1.5	5	<20	10	0.71	140	
15			SS550	1	10	<20	10	0.88	180	
18										
19										
	END OF HOLE									



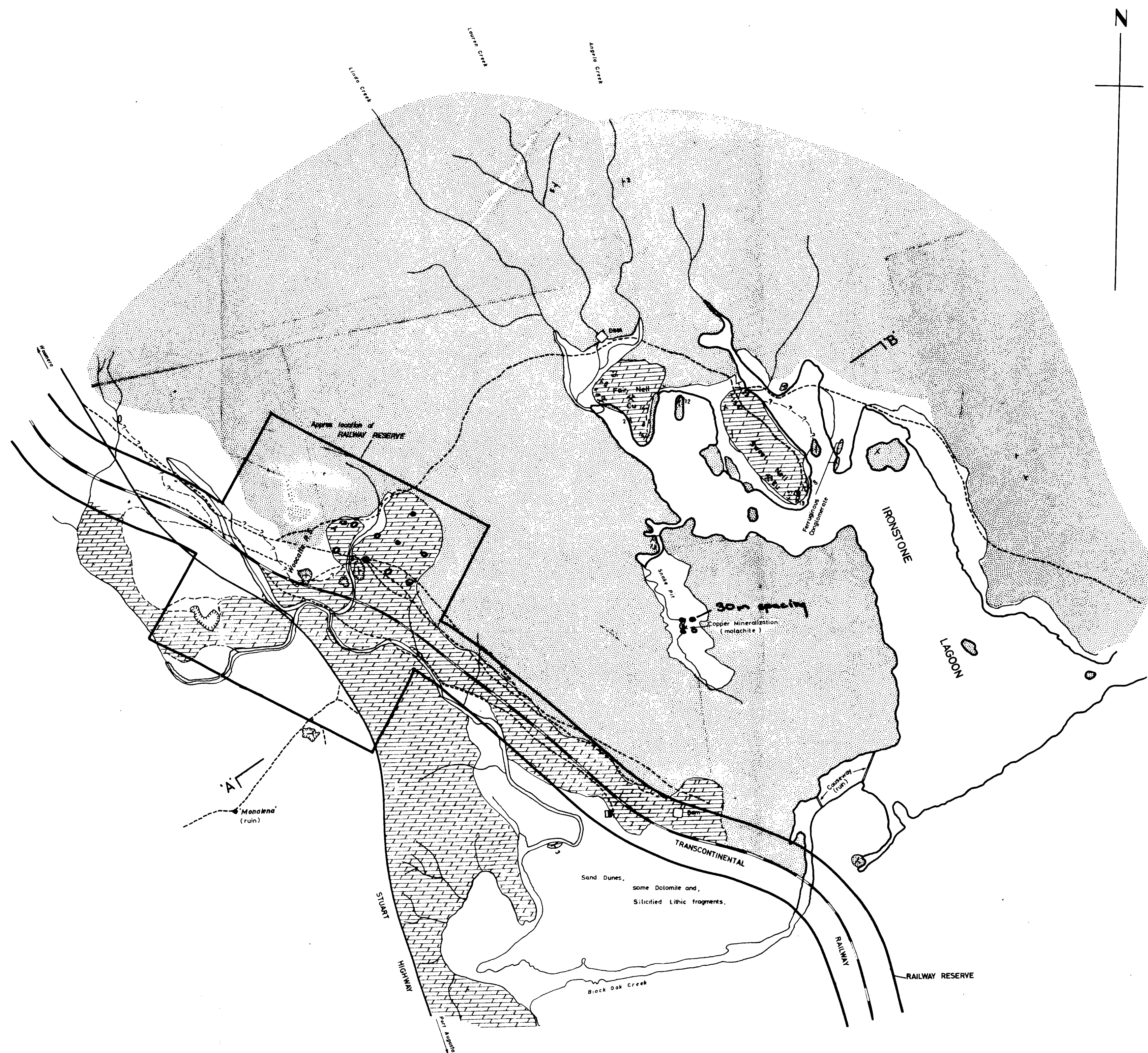
Outline of Dolomite outcrop

NOTE:
Arbitrary grid only

MOUNT GUNSON MINES PTY. LTD.
SWEET NELL MINES
LOCATION of DRILL HOLES
SCALE 1:1000metres
GEOLOGY: P.C.
SURVEYED: JCW B.M.K.
DRAWN: RXY
18 FEB 1971

DWG No 94

6626-1

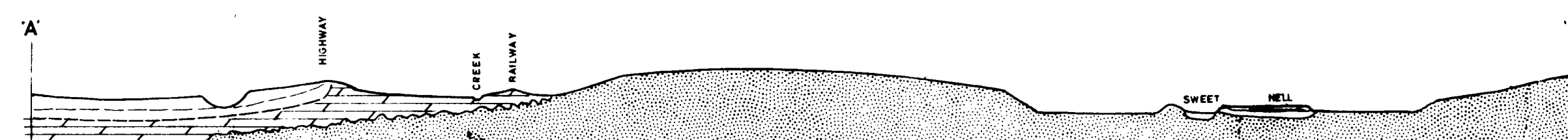


NOTE:
 ① Base map uncontrolled.
 ② Areas of surface ferruginization have not been included.

- Gypseous muds and silts
- River alluvium
- Quaternary Sands
- WHYALLA SANDSTONE Outcrop
- WHYALLA SANDSTONE Float
- WOOCALLA DOLOMITE Dolomitic clay
- WOOCALLA DOLOMITE Stromatolitic dolomite - outcrop
- WOOCALLA DOLOMITE Stromatolitic dolomite - float
- WOOCALLA DOLOMITE Dolomitic slate
- PANDURRA FORMATION Outcrop
- PANDURRA FORMATION Float
- Unconformity

- LITHOLOGICAL BOUNDARIES
- Established
 - Approximate
 - Inferred
- Main Road
- Track
- Creek
- Railway
- House
- Shaft
- Magazine
- Quarry
- Mine
- Copper
- Dip of bedding

• ~~REMOVED~~ DRILL HOLES.

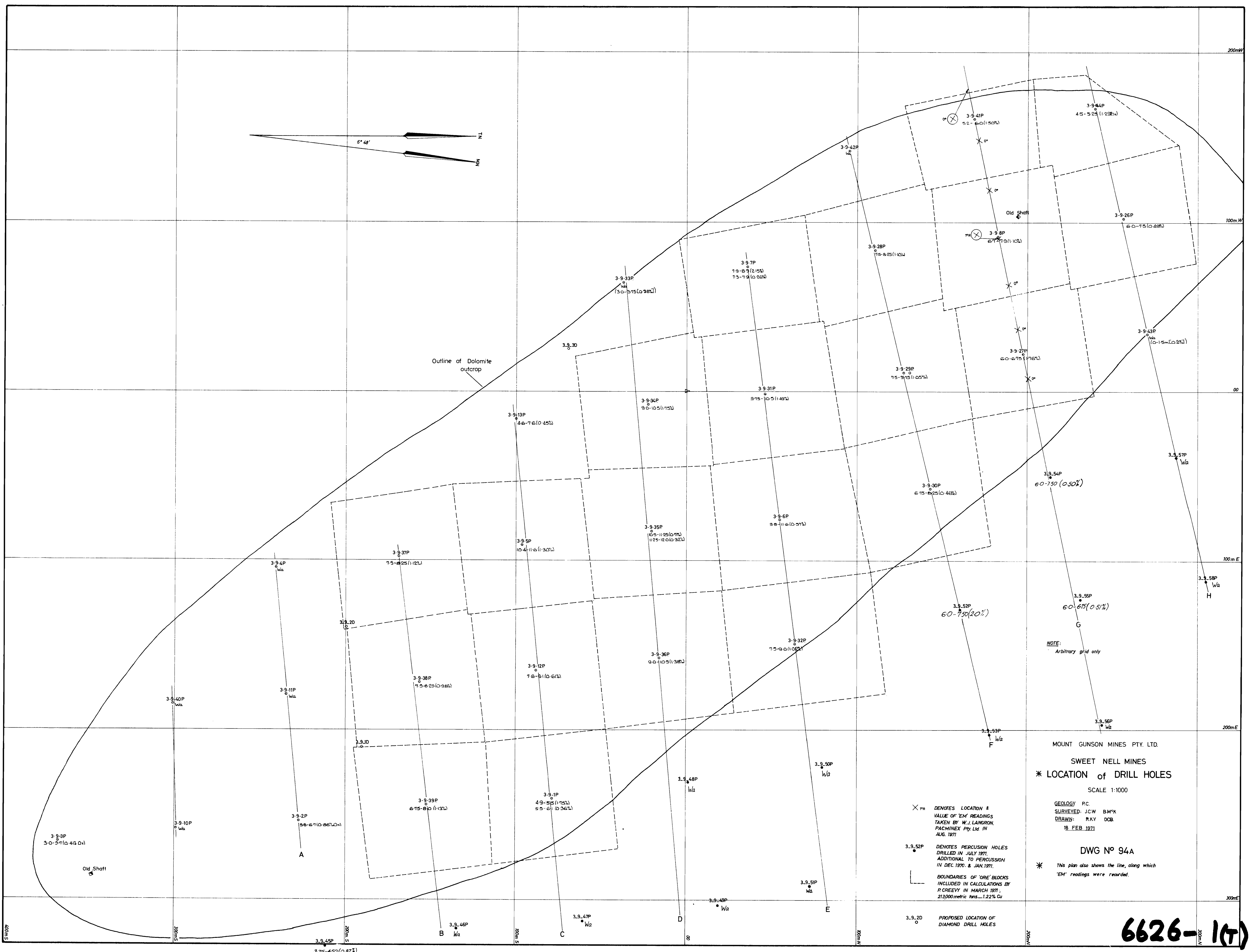


DIAGRAMATIC CROSS SECTION A'-B'

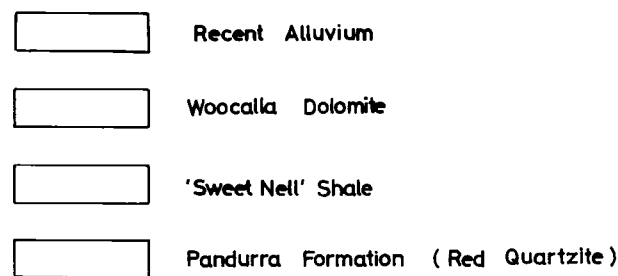


MOUNT GUNSON MINES PTY. LTD.		
WOOCALLA - IRONSTONE LAGOON AREA		
GEOLOGY		
SCALE: 1:17000 approx.	DRAWN: D.C.B.	DWG N°
SURVEYED:	CHECKED: J.M.C.	152
GEOLOGY: J.C.	DATE: 25-5-71	

6626-3

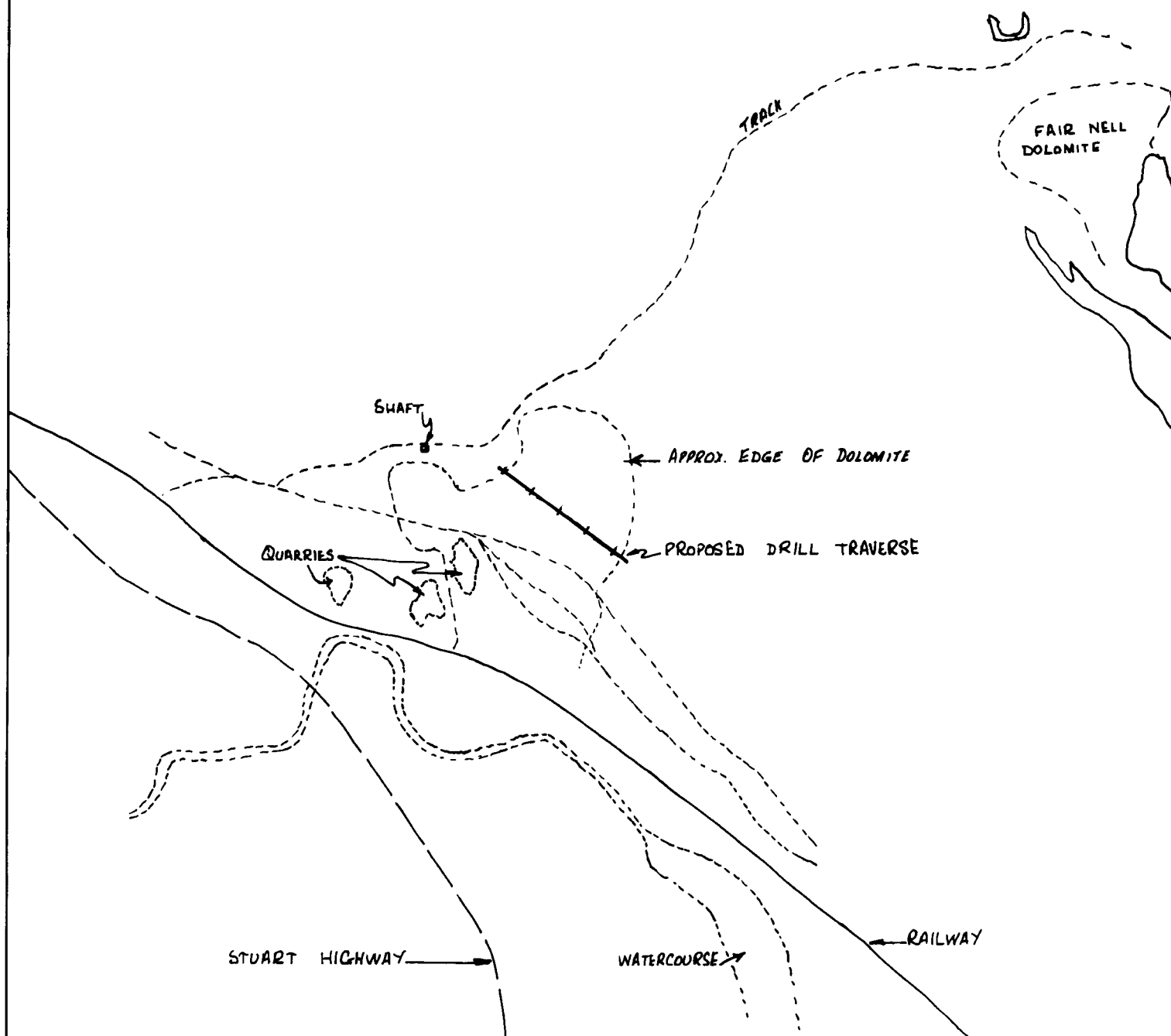
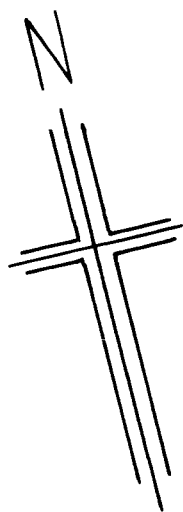


6626-1(T)



MOUNT GUNSON MINES PTY. LTD.			
SWEET NELL MINES			
GEOLOGICAL CROSS SECTIONS			
SCALE 1:1000		DWG N° A2	
SURVEYED J.C.W. B.M. 4	REVISED 18.10.71	104a	
GEOLOGY P.C.	CHECKED		
DRAWN R.K.Y. DCB	DATE 25-3-1971		

N	R.K.Y	DCB	DATE	25-3-1971
6626-2(T)				



SCALE



MOUNT GUNSON MINES PTY. LTD.

SKETCH OF PROPOSED DRILL TRAVERSE
NEAR WOODALLA DOLOMITE QUARRIES

SCALE: 1:17000 (APPROX.)

DRAWN: P.C.

DATE: 11 MAY 1971

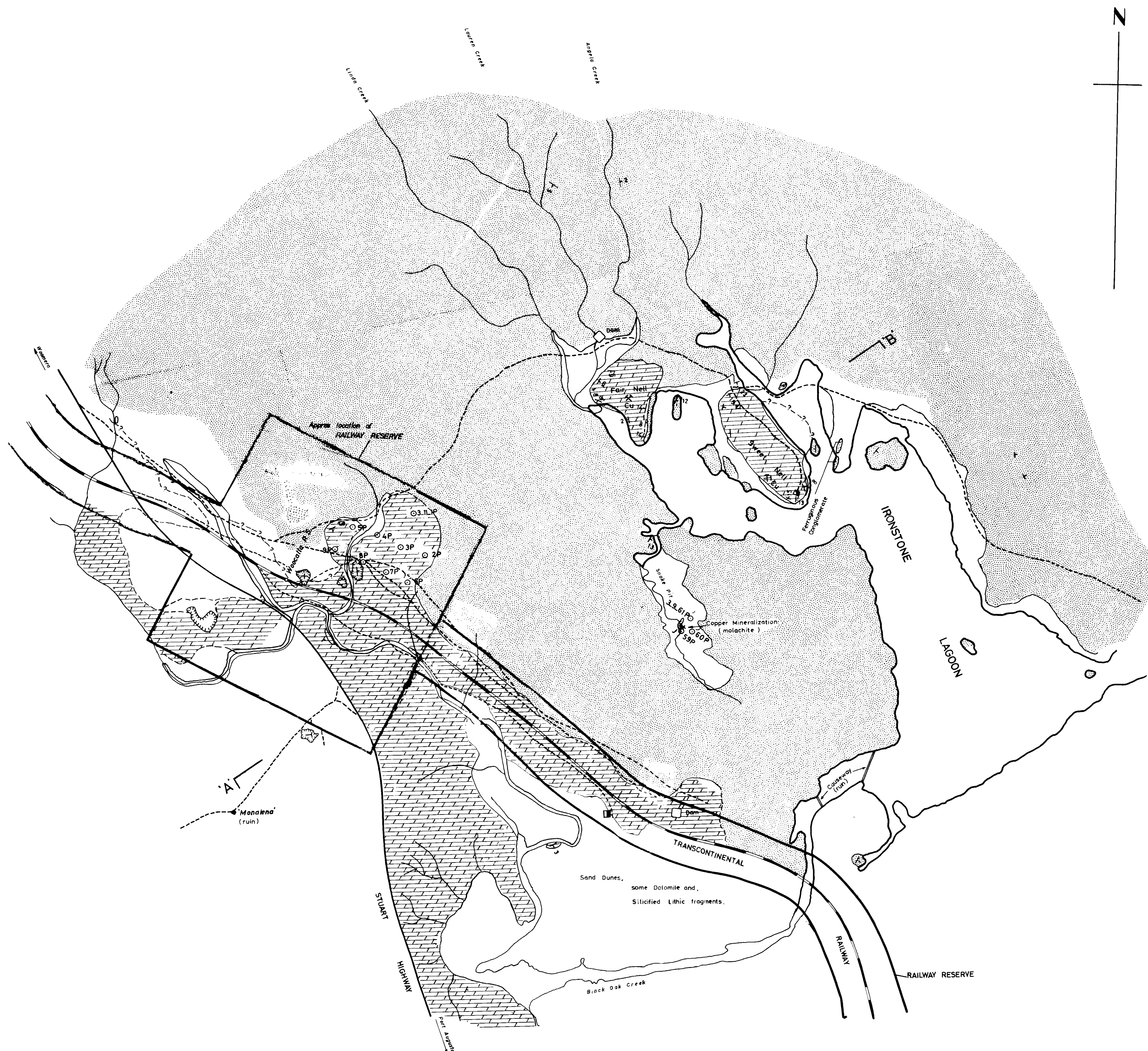
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SITE ENG.

DWG N°

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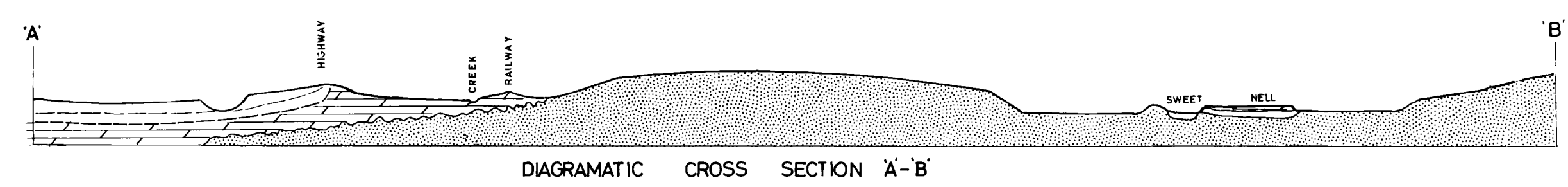


NOTE:

- ① Base map uncontrolled.
- ② Areas of surface ferruginization have not been included.

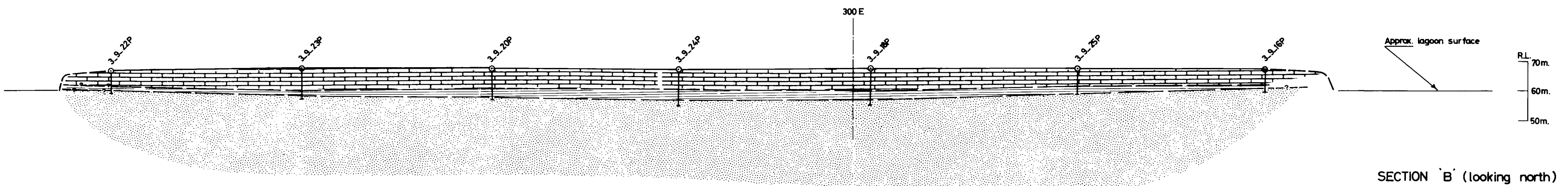
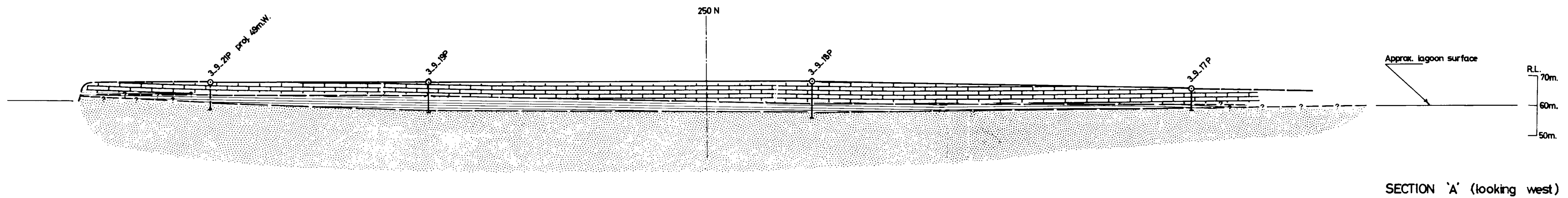
- Gypsiferous muds and silts
- River alluvium
- Quaternary Sands
- WHYALLA SANDSTONE Lithic sandstone (silicified) Outcrop
- Float
- WOOCALLA DOLOMITE Dolomitic clay
- Stromatolitic dolomite - outcrop
- " " - float
- Dolomitic shale
- PANDURRA FORMATION Red quartzite Outcrop
- Float
- Unconformity

- LITHOLOGICAL BOUNDARIES
- Established
 - Approximate
 - Inferred
 - Main Road
 - Track
 - Creek
 - Railway
 - House
 - Shaft
 - Magazine
 - Quarry
 - Mine
 - Copper
 - Dip of bedding



MOUNT GUNSON MINES PTY LTD.			
WOOCALLA - IRONSTONE LAGOON AREA			
GEOLOGY			
SCALE: 1:17000 approx.	DRAWN: D.C.B.	DWG. N°	
SURVEYED: J.C.	CHECKED: J.M.C.M.	152	
GEOLOGY: J.C.	DATE: 25-5-71		

6626-4(T)

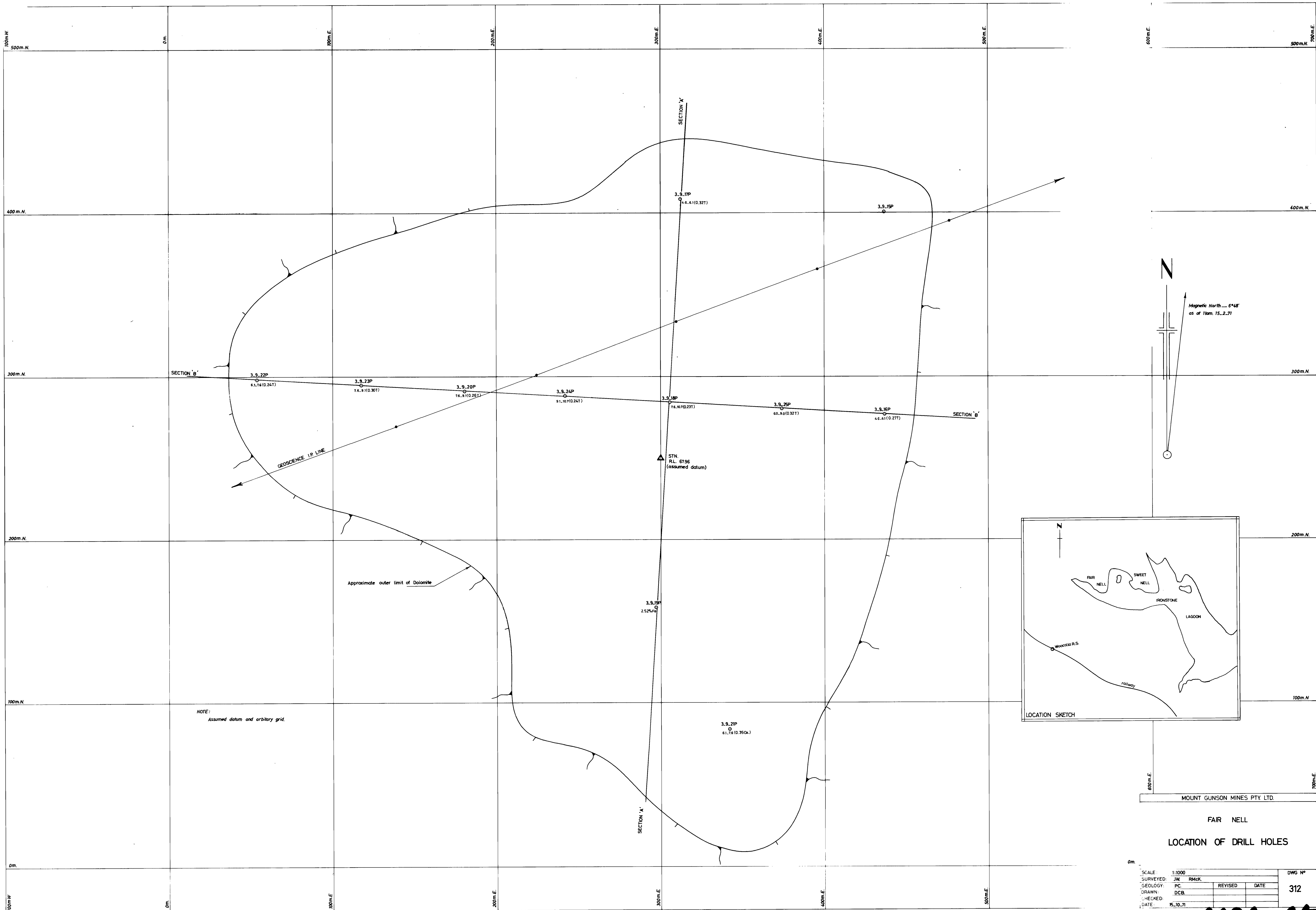


NOTE: Arbitrary grid and assumed datum.
Lagoon surface level is very approx.

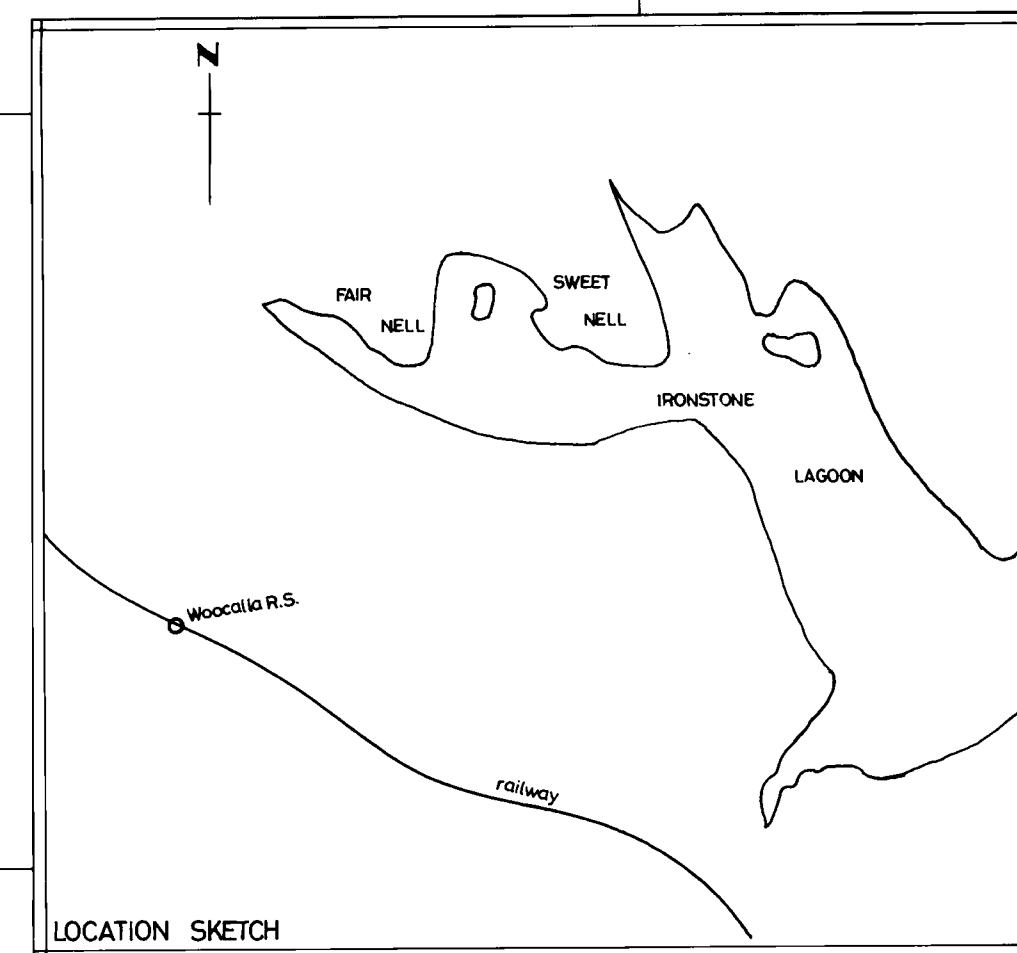
	Woocalla Dolomite
	'Sweet Nell' Shale
	Pandurra Formation

MOUNT GUNSON MINES PTY. LTD.			
FAIR NELL AREA CROSS SECTIONS			
SCALE	1:1000	DWG N°	A2
SURVEYED	R.McK.	REVISED	
GEOLOGY	P.C.	CHECKED	
DRAWN	DCB	DATE	14.10.71

6626-5(T)



NOTE:
Assumed datum and arbitrary grid.

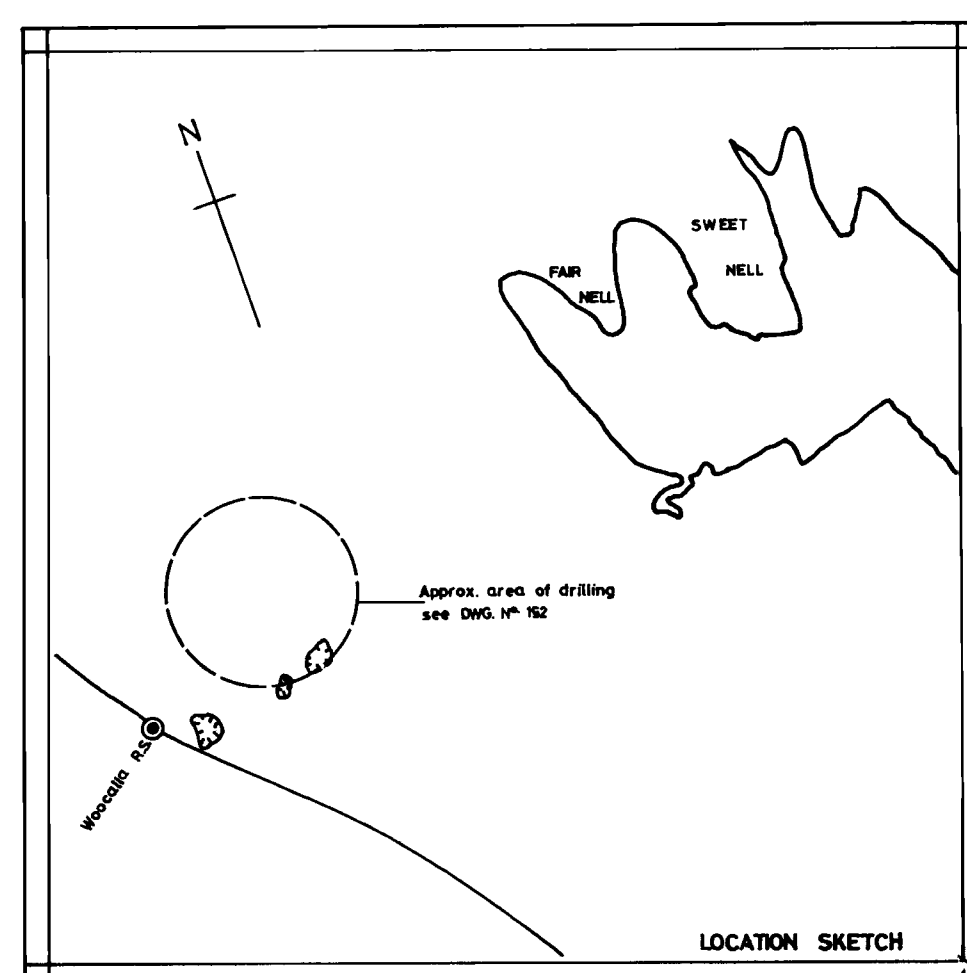
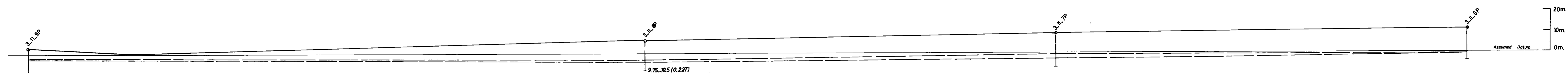
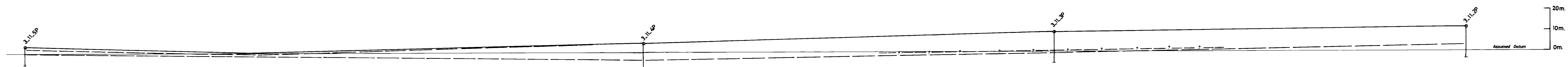


MOUNT GUNSON MINES PTY. LTD.

FAIR NELL
LOCATION OF DRILL HOLES

SCALE:	1:1000	DWG N°
SURVEYED:	JW RMck	312
GEOLOGY:	PC	REVISED
DRAWN:	DCB	DATE
CHECKED:		
DATE:	15.10.71	

6626-6(T)



NOTE: All sections drawn looking North East.

- ☐ Recent Alluvium
- ☐ Woolcalla Dolomite
- ☐ 'Sweet Nell' Shale
- ☐ Pandurra Formation (Red Quartzite)

MOUNT GUNSON MINES PTY. LTD.		
WOOCALLA		
CROSS SECTIONS		
(along drill traverses to N.E. of quarries)		
SCALE: 1:1000	DRAWN: OCB.	DWG N° 314
SURVEYED:	CHECKED:	
REFERENCE: DWG. N° 152	DATE: 19.10.71	

6626-7(T)



MOUNT GUNSON MINES PTY. LTD.		
IRONSTONE LAGOON		
GEOCHEMICAL MUD SAMPLING		
SCALE: 1:5000	GEOLOGY: RG	DWG. NO.
SURVEYED:	DRAWN: BKY	393
REVISED:	CHECKED:	DATE: 26/5/12

6626-9(7)

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Preliminary
Copy

MT. GUNSON, PERNATTY LAGOON
GEOLOGICAL MAPPING REPORTS.

PACMINEX PTY LTD

Dated in Sydney
11/8/71

Prepared by: J. Chisholm

CONTENTS

1. INTRODUCTION TO REGIONAL STRATIGRAPHY
2. ELIZABETH AREA GEOLOGICAL MAP REPORT
3. COLANDRA-MANGANESE AREA GEOLOGICAL MAP REPORT
4. WOOCALLA-IRONSTONE LAGOON AREA GEOLOGICAL MAP REPORT
5. PERNATTY LAGOON BIBLIOGRAPHY (PUBLISHED).

000089

AN INTRODUCTION TO THE REGIONAL
GEOLOGY OF THE PERNATTY LAGOON AREA, S.A.

STRATIGRAPHIC NOMENCLATURE

The sub-division of the Australian Proerozoic is now widely accepted and consists of three time-rock units i.e. (Adelaidean, Carpentarian and "Nullaginian" system) with the boundaries being approximately 600 - 1400 my, 1400 - 1800 my and 1800 - 2300 my respectively. Adelaidean is of prime interest as the sediments in the Pernatty Lagoon area are of this age. Further sub-division of the Adelaidean system is possible by two different methods. The first and oldest sub-division is into four series i.e. (Willouran, Torrensian, Strutian and Marinoan) which is based on a time-rock basis. The second method is on a rock basis by grouping the sediments within the Adelaidean geosyncline into four rock (lithological) groups. Both methods of sub-division with their various relationships are shown in Fig. 1.

REGIONAL SETTING

The lithologies that host the copper mineralisation at Mount Gunson were deposited on the Stuart Shelf during the Adelaidean Period. The Stuart Shelf is a tectonic feature that is bounded in the west by the older middle Proterozoic Gawler Platform and in the east by the Adelaide Geosyncline.

Regionally the oldest rocks are the Carpentarian Schists and gneisses (Cleve Metamorphics) that make up the crystalline basement of the Gawler Platform. (See Fig.2). Following the deposition, folding and metamorphism of the Carpentarian sediments there followed a period of prolonged and deep subaerial plantation that reduced the ancient continent to near base level. This plantation was effected under desert conditions and accentuated by the lack of vegetation cover.

The radical change in palaeogeography between the Carpentarian and Adelaidean (Fig. 3) was brought about by wide spread, spirogenic down-warping along a central longitudinal zone that stretched from Adelaide through to Arnhemland. Within South Australia this down-

warping initiated the development of the Adelaide geosyncline and commenced sedimentation that was to continue almost continually until Cambrian time. In the west the detrital material was derived from the Gawler Platform and deposited on the Stuart Shelf (marginal to the geosyncline) with pronounced thickening of the sediments in the geosyncline proper further to the east.

The first sediments to be deposited in the Adelaidean were the Callanna Beds, and their equivalents, during the Willouran series, and lies unconformably on the crystalline basement with a thickness of up to 3600 metres comprising of siltstones, dolomites and quartzites. Within the Callanna Beds there occurs a layer of volcanics (Roopena and Wooltana Volcanics) which are essentially a sequence of trachytes, andesites and basalts which is not unexpected in the early geosynclinal stage of a tectonic cycle.

The upper Willouran saw the first sediments being deposited in the Pernatty Lagoon area as evident by the Pandurra Formation which is an arenite sequence within the Upper Callanna Beds. These can be correlated with the River Wakefield Group further to the south east. Although there was continuous sedimentation within the geosyncline during the Torrensian and Sturtian series there occurred a break on the Stuart Shelf which resulted in a disconformity between the Pandurra Formation and the overlying Marinoan Strata. The Marinoan sediments are represented in the Pernatty Lagoon area by the Wilpena Group of which the Tent Hill Formation is the most important. The accompanying cross-section shows the relationship of the sediments to the Shelf and geosyncline.

C. STRATIGRAPHIC EQUIVALENTS

WILPENA GROUP

- Tent Hill Formation - ABC Range Quartzites
 - Brachina Formation
 - Ulupa Siltstones
 - Nuccaleena Formation

UPPER CALLANNA BEDS

- Pandurra Formation - River Wakefield Group
 - Pindyin Sandstone

D. STRATIGRAPHIC AGE

The lack of rocks suitable for radio-isotope dating has meant that very few definite ages can be assigned to the sediments in the Adelaide geosyncline. An age dating of rocks from the Mt. Painter and Olary districts puts the age of the crystalline basement at 1600 my and thus sets an upper limit. The base of the Adelaidean System is defined by the base of the Callanna Beds at a locality near Wooltana but unfortunately no isotope dates are available as the base of these beds defines the top of the Carpentarian System. Some indication of exact age can be obtained from isotope dating carried out on the volcanic sequence. The Wooltana Volcanics yield an age of 850 my which is considered unsatisfactory but the Roopena Volcanics (geographically separated from the Wooltana Volcanics but correlated by Thompson) yield a date of 1345 [±] 30 my.

Alternatively the time series can be used whereby the Pandurra Formation is of Upper Willouran series in age and the Tent Hill Formation considered to be upper Middle Marinoan series in age.

E. LOCAL MOUNT GUNSON STRATIGRAPHY

	Simmens Quartzite
Tent Hill	Carraberra Sandstone
Formation	Tregolana Shale
	Whyalla Sandstone
	Disconformity
Woocalla	Woocalla Dolomite
Dolomite	Disconformity
Pandurra	
Formation	Red Quartzite

1. Pandurra Formation

The clastic sediments of this formation are the oldest rocks occurring in the area and consist of hard, impervious sandstones, grits and quartzites with greywackes and conglomerates occurring locally. Typical of these sediments are the red colour of contorted "liesgang" rings formed by rythmic precipitation of iron minerals now represented by goethite and hematite. These cross-bedded, shallow water, continentally deposited arenites lie unconformably above the Gawler Range and Roopena Volcanics and thin over the Pernatty Culmination with thickening to the east and west. Thicknesses of 500 feet and 500 feet+ have been recorded in the Woomera No. 1 and Beda Bores respectively.

2. Woocalla Dolomite

This sequence of dolomite, dolomitic shales and shales unconformably overlies the Pandurra Formation and indicates the former strand line. The dolomite represents an ancient reef built from colonies of collenia with associated reef breccias and stromatolitic dolomites. Shallow water features such as cross-bedding, oolites, breccias, lode costs and sandy interbeds are common.

Thicknesses of 400 ft, 300 ft and 100+ft have been recorded in the Woomera No. 1, Beda and Maslin Bores respectively. The age of this unit is in doubt as no definite correlations can be made but it is thought to be of Lower Marinoan age.

3. Whyalla Sandstone

This arenite unit combined with the unconformably Pandurra Formation comprised the Pernatty Grit of Johns (1968). Consisting of porous lithic sandstones, local conglomerate, shale bands and some lenses of dolomite the sequence represents a transition from continental to marginal marine sedimentation. A very characteristic feature of the sandstones is the marked rounding of the quartz grains. Thicknesses of 300 ft and 200+ft are recorded in the Woomera No. 1, and Beda Bores respectively with thinning over the Pernatty Culmination.

4. Tregolana Shale

Equivalent to the Woomera Shale and consists of a sequence of purple and brown shales and siltstones which seldom outcrop. A thickness of 600 ft is recorded in the Woomera No. 1 Bore.

THE ADELAIDE SYSTEM

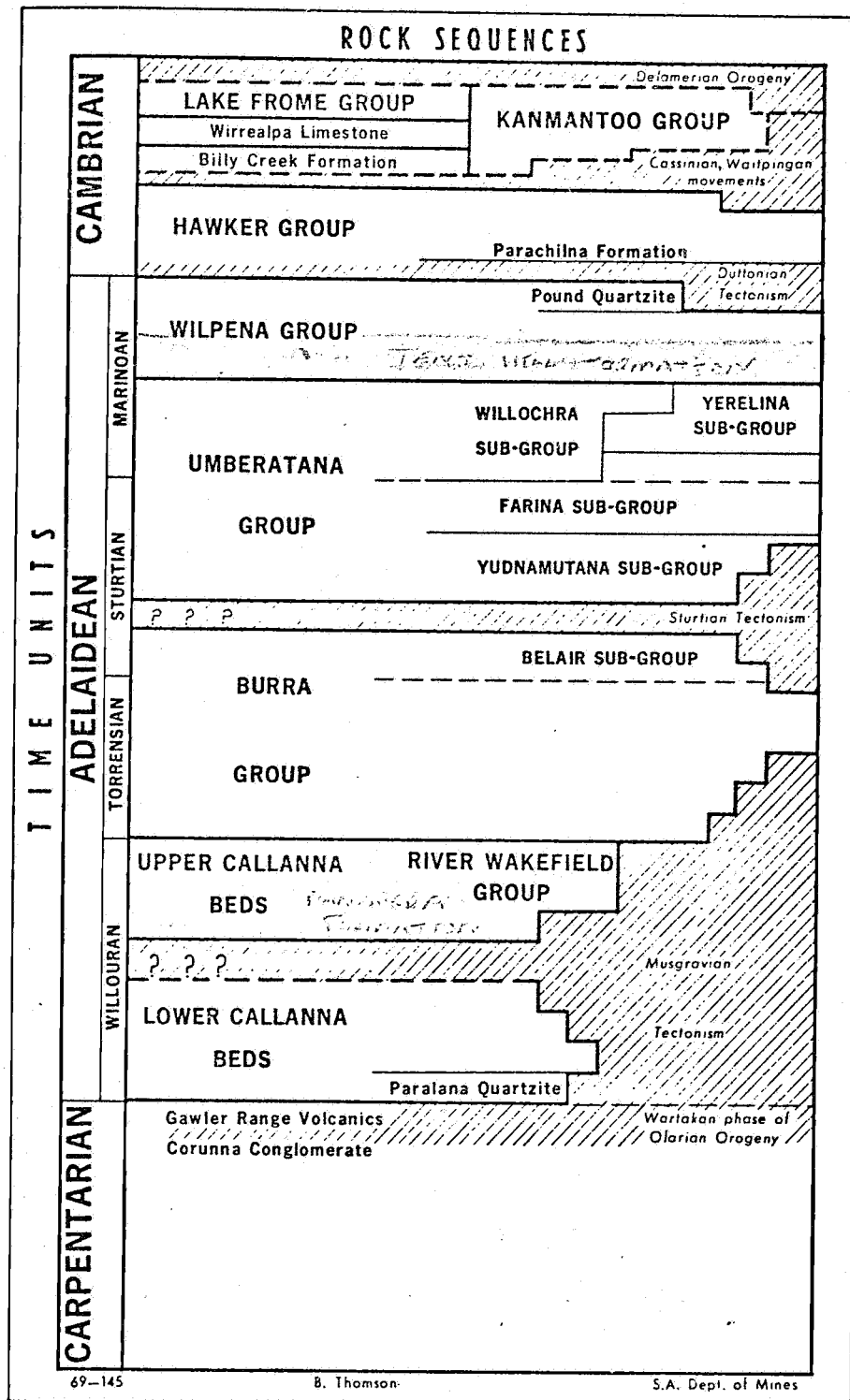
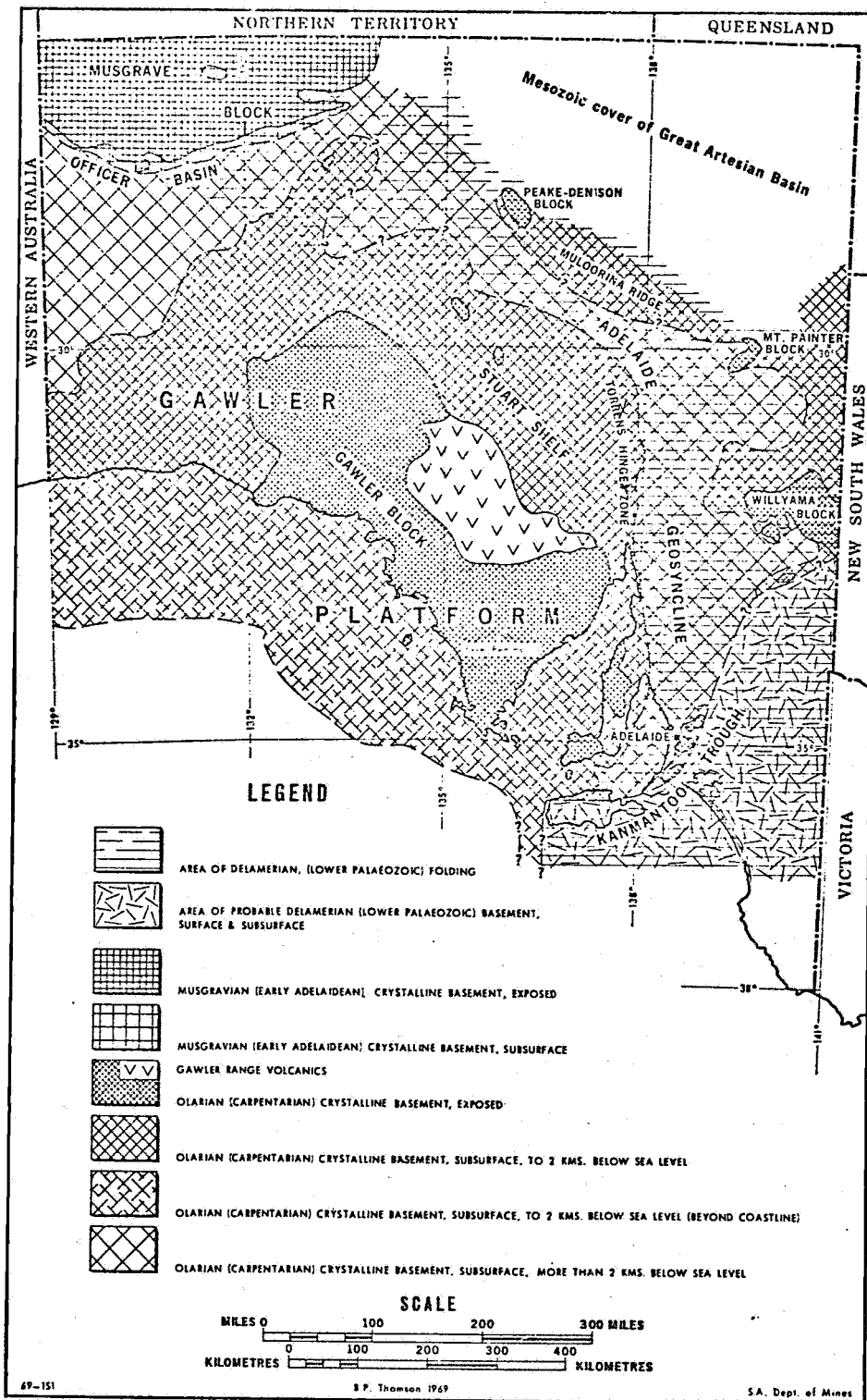


FIG-16: Time and rock terms used in the Adelaide Geosyncline and adjoining areas.

Fig 1

SOUTH AUSTRALIAN GEOLOGY



-FIG. 4- Major structural units of South Australia.

Fig 2

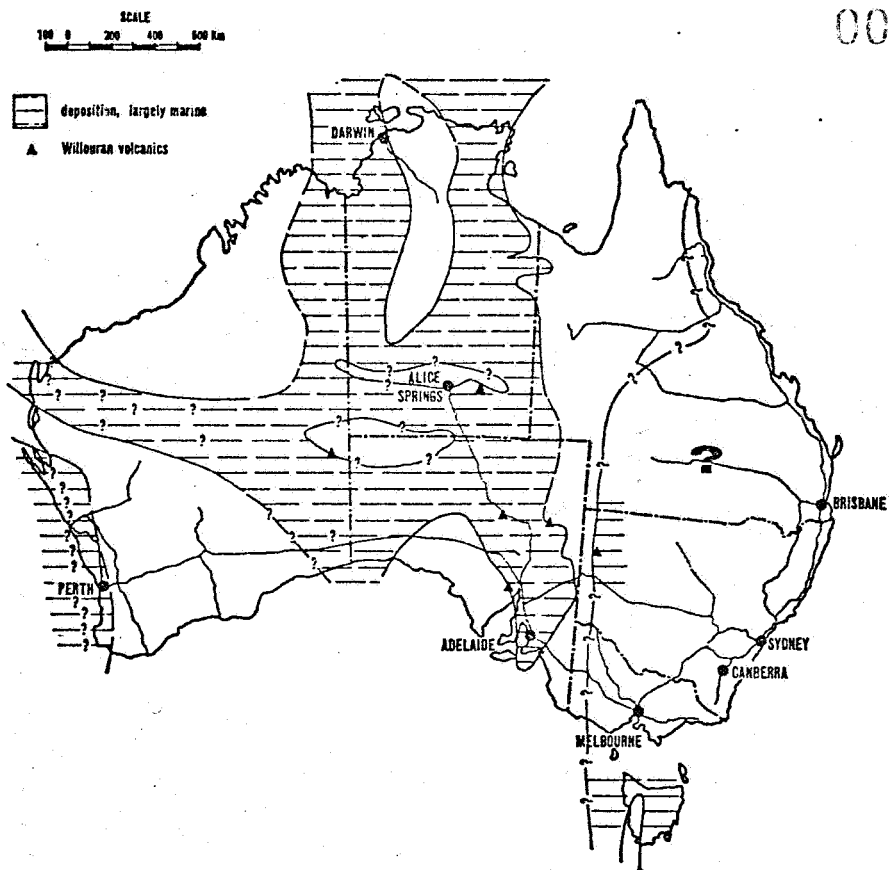


FIG. 1.13. Adelaidean palaeogeography.

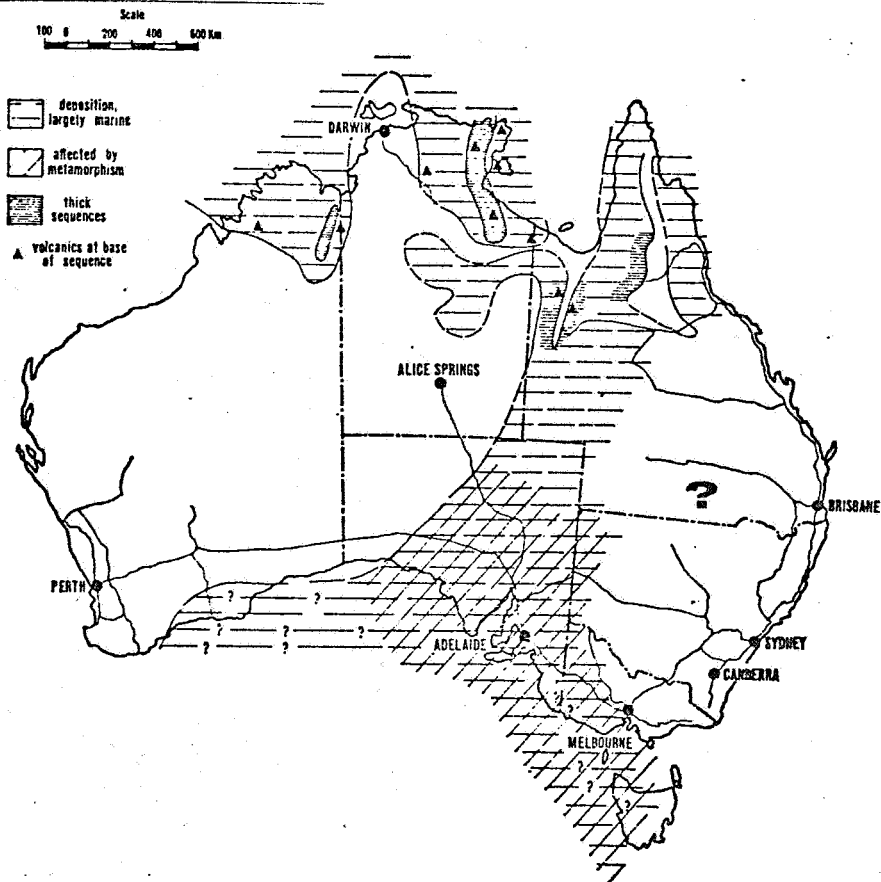


FIG. 1.12. Carpentarian palaeogeography.

Fig 3

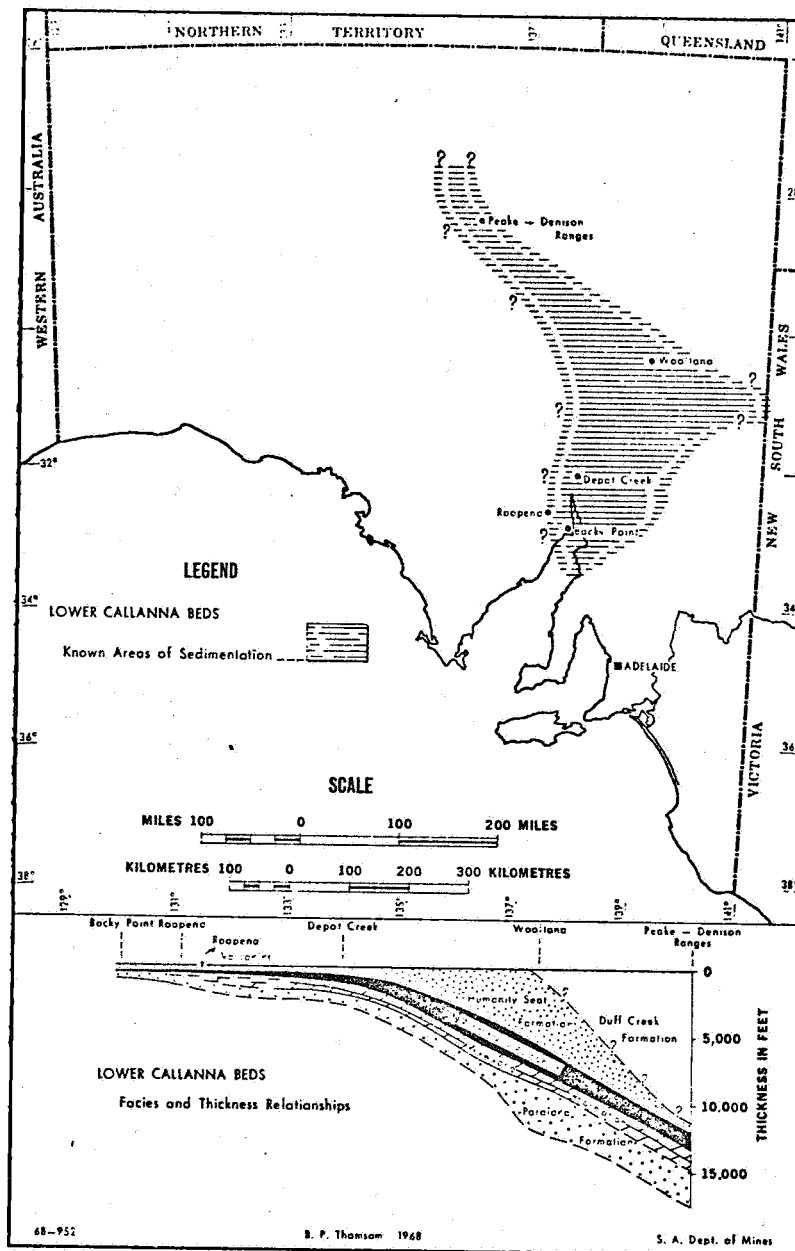


FIG. 17. Sedimentation and facies relationships of the Lower Callanna Beds.

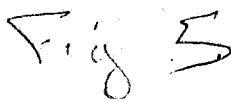


FIG. 19. Areas of sedimentation, and diagrammatic section showing Upper Callanna Beds, River Wakefield Group and Burra Group.

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THE ADELAIDE SYSTEM

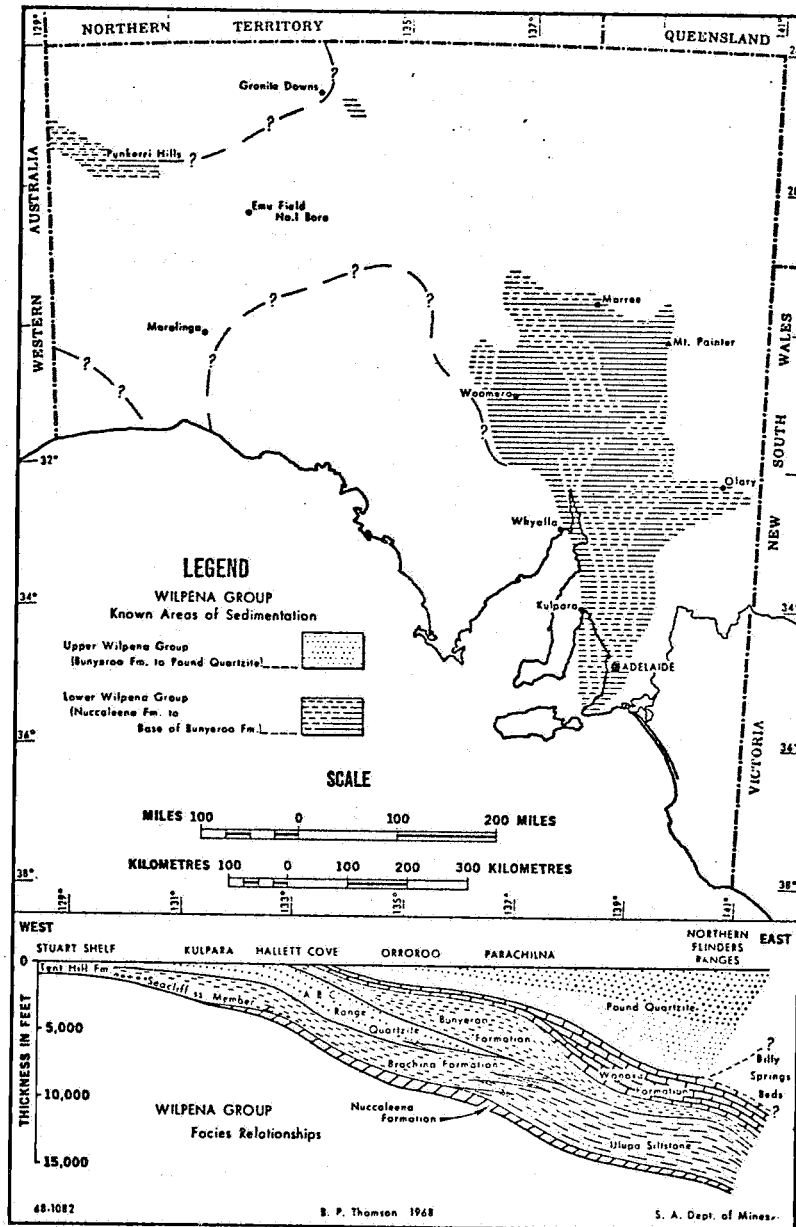


FIG. 27. Areas of sedimentation, and diagrammatic section showing facies relationships of the Wilpena Group.

Fig. 6

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REPORT ON GEOLOGIC MAPPING
ELIZABETH AREA, MT. GUNSON S.A.

PACMINEX PTY LTD

Dated in Sydney
10/8/71

Prepared by: J.M. Chisholm

CONTENTSPAGE

INTRODUCTION	1
STRATIGRAPHY	1
LITHOLOGY	4
SEDIMENTATION	6
STRUCTURE	6
ECONOMIC POTENTIAL	7

MAPS

Fig. 1 - Mt. Gunson - Elizabeth Area

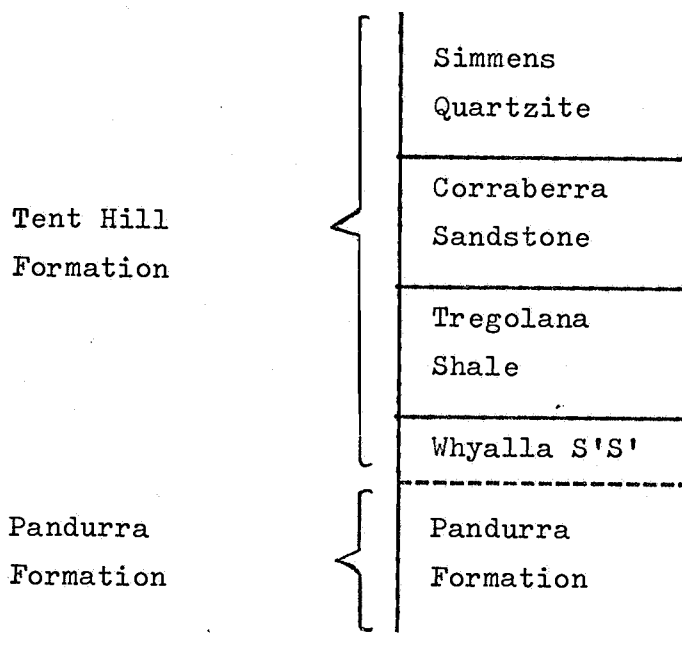
INTRODUCTION

The area covered in the mapping programme extends from the Elizabeth River, north of Pernatty Lagoon to the Western section of Bonney Bluff. Most of the area is covered by dune sand but sufficient outcrop is available for mapping. Silicification of the various lithologies makes it very difficult to identify the rock in hand specimen. In the vicinity of Bonney Bluff the stratigraphic relationships are masked by transported debris and erosional remnants.

The main aim of the programme was to test the possibility of the area being a suitable host environment for copper mineralisation. The exploration hypothesis used was that the copper mineralisation occurs in tectonic and/or topographic depressions in the Pandurra Formation.

STRATIGRAPHY

With the exception of the Woocalla Dolomite there is a complete stratigraphic sequence from the Red Quartzites of the Pandurra Formation through to the orthoquartzites of the Simmens Quartzite Member. The stratigraphic relationships are shown below:-



As mentioned earlier, there were complications due to silicification and transported debris which caused uncertainty in mapping the stratigraphic sequence. The boundary between the Tregolana Shale member and the overlying Corraberra Sandstone Member is transitional and is arbitrarily placed at a boundary where the sand content decreases below 25%. Where silification masks the grain sizes, and relative proportions of silt and sand it is impossible to ascertain the identity of the rock unit. The more resistant nature of the silicified lithologies has caused them to remain as erosion relics after the softer siltstones and mudstones have been removed and it is common to find areas of younger rock that appear to be lower stratigraphically than older less resistant strata. This is caused by less resistant sediments being preferentially eroded from below a resistant bed which remains after the underlying bed is removed. The prime example of this is at Bonney Bluff where flaggy quartzites of the Simmens Quartzite Member appear to occur below the Corraberra Sandstone Member in the form of an apron around the Bluff.

Pandurra Formation - Considerable lithological variation occurs within this formation with the occurrence of friable, coarse sandstones, gritty sandstones, pebbly sandstones and boulder conglomerates. Two units were delineated on the basis of colour and grain size. The lower unit is the characteristic red, coarse grained often gritty quartzite which comprises most of the Pandurra Formation. Conformably above the red coloured quartzite there occurs a white-buff-purple quartzite that is in part a friable sandstone. The white and buff quartzites are medium to coarse grained but generally equigranular within any bed. A boulder conglomerate has been included in this unit although the exact relationship is uncertain. There is a possibility that the conglomerate is of fluvial origin much younger than the other Pandurra Formations on account of its distribution. Only boulders of Pandurra Formation lithologies occur as clasts in the conglomerate.

11.15
dtd

Whyalla Sandstone Member - A thin strata of heavily Mn-Fe indurated lithic sandstone occurs unconformably above the Pandurra Formation. Rounding of the grains is exceptional and at the base there occur angular clasts of red quartzite which are typically not subjected to Mn-Fe induration. There is an observed gentle dip towards the NE (i.e. basinwards away from the ancient land mass) existing during deposition of this unit).

Tregolana Shale Member - The boundary between the underlying Whyalla Sandstone is not seen in outcrop but is seen to be conformable in the Gunyah Lake area to the south. The upper boundary with the Corraberra Sandstone is transitional which causes difficulty in field identification. This member is composed of a sequence of mudstones, siltstones and silty sandstones which show a wide variation in colour. Generally the mudstones are green, red and purple with white coloration where weathering has bleached the sediments. Fine grained sandstone interbeds occur within the sequence and these show considerable contortion (ball and pillow structures, load casts) due to plastic deformation of the sediments shortly after deposition (early diagenesis).

In one outcrop the argillaceous sediments are folded due to slumping. The Tregolana Shale Member is equivalent to the Woomera Shale Member of Johns (1968).

Corraberra Sandstone Member - Conformably overlies the Tregolana Shale with a transitional boundary. This sequence consists of siltstones, sandy siltstones, sandstones and quartzites that are frequently cross-bedded. The sediments are flat lying and pass conformably into the overlying sandstones and quartzites of the Simmens Quartzite Member.

Simmens Quartzite Member - Equivalent to the Arcoona Quartzite (Johns 1968) and forms the capping for the residual remnants of the Arcoona Plateau. In outcrop the quartzites appear as flaggy blocks that are very resistant to weathering and give a false impression as to the nature and affinity of the underlying bedrock.

In the vicinity of Pernatty Lagoon the only outcrops of Simmens Quartzite are on Mt. Gunson and Bonney Bluff.

Land Surfaces - Three land surfaces are evident and they are:-

- (1) Present day land surfaces which show medium relief.
- (2) Tertiary land surface that shows up as a flat lying plane over which there is a marked silicification with the result that the land surface is now represented by remnant mesas. At one outcrop there occurred a fossilised pisolitic laterite profile which indicates that the climate in the area was probably once tropical-sub tropical. The laterite development is up to 4-5 feet in thickness with rounded grains of hematite (considerable size variation) cemented by a limonite - jarosite stained silica.
- (3) Mesozoic land surface which is found at the top of the ancient Arcoona Plateau and is approximately 800 - 900 feet above sea level.

LITHOLOGY

Pandurra Formation - Composed primarily of subangular to sub-rounded quartzite grains with evidence of some geldspar minerals having been present. Any feldspar that was present has been heavily altered with only remnants of clay minerals remaining to fill the voids. The clasts that occur in the grits and quartzites indicate the nature of the distributive province and are mainly acid volcanics, chert, jasper and milky quartz. The composition of the boulders in the conglomerate are different in that they consist of older red quartzite. Silicification has masked the grain sizes and replaced the caly minerals to give the rocks the appearance of clean, quartz-rich quartzites.

Whyalla Sandstone - The blocks of red quartzite (Pandurra Formation) that occur at the base of the sandstone have been silicified and exhibit Liesegang rings whereas the sandstone does not. The sandstone consists of extremely well rounded quartz grains in which the degree of sorting is well advanced. With the silicified quartzite below and a mudstone above the Whyalla sandstone has acted as an aquifer to aqueous solutions. This is evident by the heavy iron and manganese staining throughout the exposed outcrop.

Tregolana Shale - This sequence of sediments passes from fine grained laminated mudstones through interbedded sandstones and mudstones to a silty sandstone. Close examination of the mudstone sequence shows that fine-grained, sandstone layers, which contain both well rounded and angular quartz grains, occur within the laminations of fine grained argillaceous material. Within the argillaceous material there can be seen both micro-convolutions and micro-faults. The sandstone interbeds towards the top of the sequence have a bimodal distribution of grain sizes with an argillaceous matrix filling voids between the well rounded quartz grains. The argillaceous material is thought to be due to diagenetic remobilisation of material following deposition.

Transition - The boundary between the Tregolana Shale and the overlying Corraberra Sandstone is fixed at 25% sand content but the variability of the sediments makes this boundary very transitory. A close examination of samples selected from this transition some showed a silicified silty sandstone containing well rounded grains, in one case angular, with 5 - 10% argillaceous material.

Corraberra Sandstone - This sequence consists of a micaceous quartzite at the base above which is a purple sandy siltstone with a micaceous, crossbedded, purple sandstone at the top. The lower micaceous sandstone is a fine grained quartz sandstone interbedded with siltstone. The angular grains are close packed with some sericite to give a clean sandstone. Above this is a purple even grained quartz-sericite siltstone interbedded with a fine grained sandstone. The

upper purple sandstone contains medium grained quartz, muscovite, sericite and iron stained argillaceous matter derived through the breakdown of feldspar minerals.

Simmens Quartzite - Generally cream and white fine grained orthoquartzites in which the grains are well rounded but silicification tends to mask the texture. Some heavy minerals (possibly tourmaline) occur and a small amount of argillaceous material occurs interstitially. Interbeds of quartzite which contain significant feldspar remnants occur. Generally the Simmens Quartzite is better sorted, finer grained, more rounded, contains no mica and has less argillaceous material than the Corraberra Sandstone.

SEDIMENTATION

The pattern of sedimentation is the result of a marine transgression followed by a partial regression. Following deposition of the Pandurra Formation there occurred movement which terminated sedimentation and initial erosion. Deposition of the eroded Pandurra clastic material resulted in the Whyalla Sandstone and as the transgression progressed these changed to mudstones and siltstones of the Tregolana Shales. From the upper part of the Tregolana Shales there was a trend in which the sediments became better sorted and rounding improved as the erosion cycle reached maturity until the clean orthoquartzites of the Simmens Quartzite were deposited.

STRUCTURE

Two structural events are evident but neither is thought to be major. One includes the lithologies of the Pandurra Formation in which slight folding has undulated the strata thus producing variations in dip and strike reading. There is however a regional dip for the Pandurra Formation. Shearing is common and minor faulting occurs with the dips of strata disrupted adjacent to the faults being up to 80°. The second structural event included all of the post-Pandurra lithologies. There is a regional dip to the SE resulting from onlap of

of Youngs Formation on to the Pandurra lithologies but is not thought to exceed 5°. Within the sediments of the Tent Hill formation there are broad shallow synclinal and anticlinal structures with minor tight folds possibly caused by slumping.

ECONOMIC POTENTIAL

A working hypothesis is that the distribution of copper mineralisation is controlled by permeability of host rock, ground water conditions, structure of the depositional environment.

On this hypothesis there is only one area, that is regarded as a possible location for mineralisation in the Elizabeth area. This is in the Whyalla Sandstone that outcrop along the mouth of the Elizabeth River. The sandstones were once permeable and would have acted as an aquifer between the red quartzites and the overlying shales. As the regional dip is to the SE there would within the basin be Eh/pH conditions where any copper in solution would be precipitated out. It is unlikely that any copper would still be forming as the heavy iron - manganese formation has markedly reduced the permeability of the sandstones.

It is proposed that a few exploration holes be drilled through the Tregolana Shale into fresh Whyalla Sandstone. The location of these holes would be in the vicinity of where the Mt. Gunson-Yeltacowie road crosses the Elizabeth River as access to the area to the south east would be restricted to an airtrack and compressor.

Manganese occurs as pyrolusite and psilomelane but is of insufficient extent to be of interest. Hematite occurs in the laterite soil profile but is again very restricted in extent. Lilitic and kaolinitic clays occur in the pallid zone of the Tregolana Shales and appears to be similar in quality to that being quarried at Magazine Hill near Woocalla.

COLANDRA - MANGANESE WORKINGS AREA MAP REPORTINTRODUCTION

This area is situated some 5 miles south of the Mt. Gunson Mine Site on the western edge of Pernatty Lagoon. The occurrence of a shale horizon beneath the dolomite that is stratigraphically equivalent to copper bearing shales at Sweet Nell and Fair Nell is of interest as well as there being a natural depression in the area.

Manganese has been mined intermittently from three locations in the area with a small amount of barite. The area in which the manganese deposits occur was mapped first with the northern extension completed at a later date in order to evaluate the dolomite as a potential host for copper mineralisation.

STRATIGRAPHY

Pandurra Formation - Forms the disconformable basement upon which the later rocks lie. The regional dip of the Pandurra Formation is to the east and appears to be in the order of $5-6^{\circ}$. Relief of the ancient Pandurra land surface was considerable and in the order of 100 feet as there can occur red quartzites of the Pandurra Formation outcropping at a topographically high level than the overlying Whyalla Sandstone. This situation is brought about by the lithic sandstones having been deposited in the valleys of the Pandurra land surface while no deposition took place on the ridges (the lithic sandstone may also have been subsequently eroded from the ridges).

Woocalla Dolomite - The dolomite occurring in the area belongs to the reef facies and the various parts of the

reef can be recognised. For mapping purposes the dolomite was sub-divided into the following types :-

Stromatolitic

Collenia colonies

Sandy

Brecciated

From this it is possible to build a complete history for the deposition of the dolomite. The various units of the dolomite are not laterally extensive within the area. Generally the base of the dolomite is identified by a dolomitic shale - mudstone, dolomite breccia, sandy dolomite or boulder conglomerate. It is thought that the shale development took place in restricted depressions on the Pandurra land surface and increases in thickness away from the shore. The shoreward facies equivalent of this shale is the boulder conglomerate that occurs along the contact between the red quartzite in a dolomite matrix but there is uncertainty as to age of the matrix. At South Point (S.W. of Woocalla) a boulder conglomerate occurs with no dolomite matrix and a sharp contact between the conglomerate and the overlying dolomite. It is possible that the Colandra Area conglomerate is equivalent except that later deposited dolomitic muds filled the voids of the conglomerate. The most feasible explanation is that the conglomerate was deposited consanguineously with the dolomite during some period of torrential rainfall as the seaward trend at the base of the dolomite is conglomerate, sandy dolomite and shale. Ferruginisation is common around the base of the dolomite as iron sulphate and bicarbonate solutions tend to oxidise on contact with the dolomite.

Colonies of collenia occur on Pandurra basement highs within the dolomite and constitute the actual reef facies. Within this reef there occurs brecciated dolomite due to wave .

action and scouring of ocean currents. Rounded quartz grains are mixed with the dolomite and are due to terrigenous contamination from rivers and streams. The upper part of the dolomite sequence is mainly a stromatolitic dolomite but at the very top of the sequence it is a sandy dolomite breccia. The breccia marks the disconformable contact with the overlying Whyalla Sandstone.

Whyalla Sandstone - The lithic sandstones once formed a sheet deposit over the older Pandurra quartzites and Woocalla dolomites but now only scattered remnants exist in the Colandra-Manganese Workings Area. Silicification and ferruginisation masks variations within this member but a fresh outcrop further to the west reveals two distinct types of sandstone. The lower type contains torrent cross-bedding (festoon) of high directionality and it is composed of rounded quartz grains and angular pebbles of red quartzite. Above this is a flat lying, friable, quartzose sandstone exhibiting minor cross-bedding and containing no clasts. The lithic sandstone which outcrops near the mouth of Honeychild Creek contains a mixture of red quartzite and dolomite clasts in a well rounded quartz matrix.

Tregolana Shale - There is very little of this member outcropping but that which does shows tight synclinal and anticlinal folding. The brown, red and green mudstones and siltstones are finely laminated.

Tertiary fluvial debris - Predominantly composed of red quartzite cobbles and boulders with no matrix. The sub-rounded clasts were deposited in tertiary rivers which flowed down existing watercourses during a period of higher rainfall and temperature (laterites).

SEDIMENTATION

As the seas transgressed over the old Pandurra land surface the deposition changed from clastic to carbonate. Following a period of erosion the sedimentation reverted to clastic. The ancient shoreline of the seas from which the dolomite was precipitated is marked by a basal conglomerate breccia and sandy facies. All of the dolomite was very close to the shoreline as pebbles are frequent in the dolomite and it is unlikely that pebbles would move very far from the land mass. Further from the land there developed shales in the depressions where conditions were quiet. On the basement highs there developed colonies of collenia. Occasional periods of ocean turbulence brecciated the reefs and left an arenite matrix to the collenia. Currents cut channels in the reef which were later filled with a sandy, brecciated dolomite. There then followed a period in which sedimentation was quiet (deeper water) with stromatolitic and massive dolomites being developed. As the seas regressed the reef again came within the zone of wave action and brecciation of the reef occurred. Clastic material contaminated the brecciated dolomite and an example of this is seen where the track crosses Honeychild Creek.

Deposition of the Whyalla Sandstone was initially in an estuarine environment where conditions were turbulent, current velocity high, shallow water and high directionality. This is evident from the cross-bedding type, variance, dip and content. Following the estuarine deposition, the second type of sandstone was deposited in a marginal marine environment where two predominant current directions produced a high variance, good sorting and generally a mature sandstone. The trend was towards deeper water sedimentation as testified by the Tregolana Shales.

STRUCTURE

Structural deformation is very slight with most of that occurring being intraformational. Undulations in the dolomite and other features are due to stresses set up through lithification, compression from overburden and pressure gradients. Folding within the Tregolana Shales is very tight but this does not exclude slumping of the sediments as the course. Jointing is ubiquitous throughout the area (best developed in the dolomite) and due to regional stress fields.

In the Manganese Workings area the Post-Pandurra sediments form a trough in the Pandurra land surface and it is possible that the southern dolomite was not connected with the dolomites in the northern area. To support this is the absence of basal conglomerate in the south and absence of shale in the north. The Colandra area has no structural or topographic depression in the area mapped with the sediments having a regional easterly dip.

ECONOMIC POTENTIAL

Copper - Samples of shale (beneath Dolomite and Tregolana) that were collected and analysed were low in copper and drilling in the Manganese Workings Area revealed a quantity of shale beneath the dolomite but no significant copper. Analysis of drilling have not been available to-date. If the analysis of the recent drilling warrant continuation of drilling then it is proposed that the area to the S.E. should be investigated to outline the black shale trend. In the Colandra Area the areas to the north and north-west have revealed stream sediment geochemical anomalies and have areas in which lithic sandstone overlies the red quartzite in an environment similar to the House - Main Area.

Manganese - The occurrence of the manganese is erratic with it occurring as thin sheets of small area. Barite and fluorite occur as veins in the dolomite and occasionally in the red quartzite. The origin of this is precipitation from saline lagoon waters in shears and joints.

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Explanatory notes to accompany the geological map for the:-

WOOCALLA - IRONSTONE LAGOON AREA

J. CHISHOLM

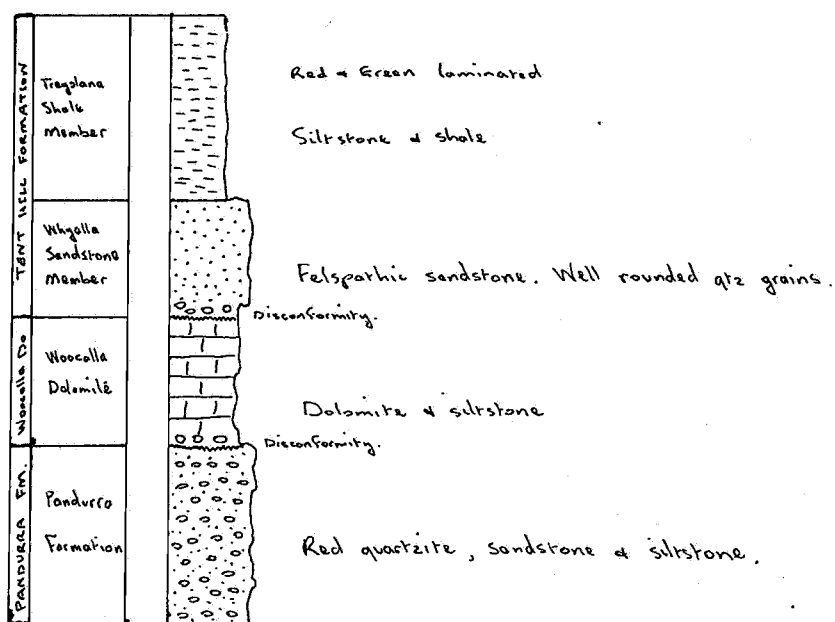
PACMINEX 1971.

INTROOUCTION:

Mapping in the Woocalla - Ironstone Lagoon area was undertaken in order to assess the economic potential. A copper bearing shale is known to exist beneath a dolomite cover at Sweet Nell and Fair Nell which was mined for malachite early in the century. The area forms a depression and it was necessary to establish whether this was topographic or tectonic and the possibility of copper mineralisation similar to East and West Lagoons Forming within. Mapping was from air photographs on a scale of approximately 1:17,000 from which an uncontrolled base map was compiled. (See Dwg. 152).

STRATIGRAPHY:

The number of stratigraphic formations and members that occur in the area are few but the facies changes within a member has caused the affinity of some lithologies to be uncertain. Basically there are only the Pandurra Formation and Woocalla Dolomite involved with the possibility of the Whyalla Sandstone and Tregolana Shale being present.



The above diagram shows the regional stratigraphic relationship of the various units. The quartzites of the Pandurra Formation are the oldest outcropping rocks in the area and form an unconformable basement upon which the younger rocks were deposited.

Following deposition of the Pandurra Formation there occurred slight tectonism which formed a tectonic basement high (Pernatty Culmination) which for a while, divided the Stuart Shelf in half.

A disconformity separates the dolomites and shales of the Woocalla Dolomite from the basement quartzites and it can be seen that the basement topography had considerable influence on the dolomite deposition. Erosion has removed some dolomite from the area but it is thought that there was never a complete cover over the area. The present distribution of outcrop is in topographic depressions within the basement topography that once constituted embayments in an ancient shoreline. Below the main dolomite unit there occurs a dolomitic shale that is copper bearing in the vicinity of the Sweet and Fair Nells. These two areas are separated from the main dolomite outcrop at Woocalla by a ridge of red quartzite. Thickness of the basal shale unit varies considerably from being absent to 3 metres. It must be mentioned that a basal conglomerate occurs beneath the dolomite in some localities (South Point) and no shale is present. It is uncertain as to whether the boulder conglomerate (boulders upto 0.6 metre in length) is a lateral facies change of the shale. The most probable explanation is that the boulder conglomerate is part of the Pandurra Formation and constitutes a stream/river deposit. This question is discussed in detail in the Manganese - Calandra area report.

The overlying dolomite is upto 10 metres in places and shows both lateral and vertical variation with respect to texture, sand content and form. Above this dolomite there occurs a shaley siltstone that outcrops in a quarry to the east of Monalena Homestead. This siltstone could be a lateral variation of the main dolomite (closer to the shoreline). A thin layer of a dololutite is found within the siltstone.

At Magazine ^{Hill} a small quarrying operation has exposed a sequence of white laminated mudstone that occurs in the pallid zone and has suffered alteration through weathering to a kaolinitic clay. This mudstone was initially thought to be a unit of the Woocalla Dolomite and has been draughted as such on the map. Following a mapping programme in the Elizabeth area where there occurs a sequence of shales, siltstones and mudstones of known Tregolana Shale affinity.

The gross similarities to these sediments are such that the Magazine Hill outcrop is assigned to the Tregolana Shale Member.

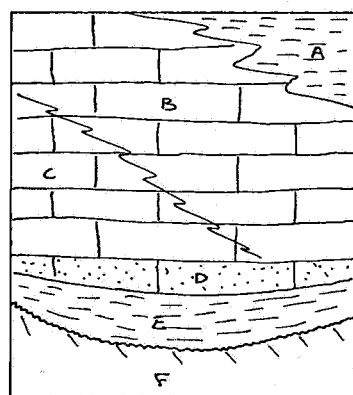
Above the mudstone there occurs some float that is identified as lithic sandstone but is assumed to have been transported and not insitu.

LITHOLOGY:

Pandurra Formation - This sequence consists of quartzites, sandstones, grits and minor conglomerate most of which show the characteristic red coloration and "liesegang ring" development. Cross - bedding indicates a source direction from the west. Within the predominantly quartzite sequence there occurs a shaley band whose surface expression is a gypseous silt. Iron staining is a common feature with some areas heavily indurated but these were generally found to die out at depth. There occurs a lineation to the east of Sweet Nell where specular hematite has indurated a line of weakness within the quartzite.

Woocalla Dolomite - This member comprises of three identifiable units as shown below.

- A Dolomitic Siltstone
- B Massive Dolomite
- C Stromatolitic Dolomite
- D Sandy Dolomite
- E Shale + mudstone
- F Pandurra Formation



The basal shale unit is dolomitic composition and acts as host to copper mineralisation. An analysis of this shale reveals quartz, dolomite, siderite, halloysite with some chalcopyrite. Above the water - table the copper is oxidised with most of the ore having been removed by mining operations early in the century. Bedding planes in the shale are accentuated by the different colours of the shale which are due to the varying oxidation states of iron.

A dolomite unit is the predominant lithology within the member and is the only lithology to be seen in outcrop. Variations exist within this unit both vertically and laterally and represent the different facies of a reef environment. At Woocalla the dolomite is extremely fine grained with little internal structure and represents a tidal mud flat. The dolomite at Sweet Nell and Fair Nell is stromatolitic in form and represents a shallow water, back reef deposition. The vertical variation is best developed at Fair Nell where the dolomite becomes sandy towards the base and in places a sandstone interbed occurs.

The upper unit occurs west of the Stuart Highway in a quarry (NE of Monalena Homestead). Quarrying has exposed a dolomitic siltstone that contain lenses of dololomite and this could represent a lateral facies variation of the Woocalla Quarry dolomite.

Whyalla Sandstone - The only occurrence of the lithic sandstone is as float on top of Magazine Hill. The sandstone occurs as a bimodal and heavily silicified sandstone. More extensive outcrops of sandstone occur further to the west and probably constitute the source of the float above Magazine Hill.

SEDIMENTATION AND STRUCTURE:

Sedimentation commenced with the fluvial deposition of the quartzites with terrigenous sands and grits coming from a source in the west. The deposition of this ^aclastic sequence was sporadic and was terminated by minor tectonism during which the "Pernatty Culmination" was formed thus changing the depositional environment.

A large break in sedimentation occurred between the ^aclastics of the Pandurra Formation and deposition of the Tent Hill Formation, arenite - dolomite sequence. The lithologies of the Woocalla Dolomite which occur in the area are the result of a shallow water, restricted (?), lagoon environment. The reef environment is discussed in more detail in the Manganese - Colandra area report where it is shown that the dolomite is in fact very close to the ancient strand line.

Following a long period of erosion of the Pandurra landsurface there occurred a transgression in which the encroaching seas invaded the valleys of the eroded topography. Carbonate deposition followed and is now represented by the Woocalla, Sweet Nell and Fair Nell outcrops which fill valleys of the Pandurra topography.

The basal shale unit formed in restricted hollows and depressions which accounts for its limited extent.

Reasons for the formation of the dolomite are possibly a fall in sea level or amelioration of the climate as either would alter the Ca/Mg concentration of the sea water to a level suitable for dolomite deposition. It is suggested that the Sweet - Fair Nell dolomite and the Woocalla Quarry were not directly joined but formed in different depressions. The Sweet - Fair Nell dolomite is a fine - grained, stromatolitic dolomite which contains some clastic interbeds. The Woocalla dolomite is clearly a shallow water variety and it can be shown that at some stage the environment was intertidal during which time the lagoon muds were periodically exposed to the sun and desiccated. The Woocalla dolomite is extremely fine grained (micritic) with numerous slump structures. Other shallow water structures exhibited are:- ripple marks, cast structures, mud cracks and clasts.

It can be said that the dolomites in the area are penecontemporaneous with the environment having been deposited in both the neritic and littoral zones as a result of deposition from hypersaline seas through either or both biochemical (algae) control of pH and direct chemical precipitation. Wow!

Structural deformation in the area is not severe and consists of a regular joint pattern in all lithologies, minor shears and intraformational slumping. Movement along joints and shears has been very small with displacement never being greater than 0.5 metres. Intraformational structure in the dolomite includes a regular concentric folding with associated décollement and boudinage structures.

MINERALOGY AND ELEMENT DISTRIBUTION:

Below the water table the copper in the shale occurs primarily as a very fine grained and disseminated chalcopyrite. Above the watertable the copper has oxidised and forms atacamite, malachite and chrysocolla.

The distribution of copper, silver, lead and zinc have been plotted for Sweet Nell and Fair Nell with the results shown in Figs. 1 - 4.

ECONOMIC POTENTIAL:

Providing that a sufficient quantity of copper bearing shale can be outlined then it may be an economic proposition. At present the only area of copper bearing shale is at Sweet and Fair Nell but a large area of dolomite occurs to the south of Woocalla and the presence of shale beneath is unknown. If a copper bearing shale occurs at one place then it is probable that it occurs at other localities and it is proposed that a programme be planned at a later stage to test the dolomite for the presence of shale.

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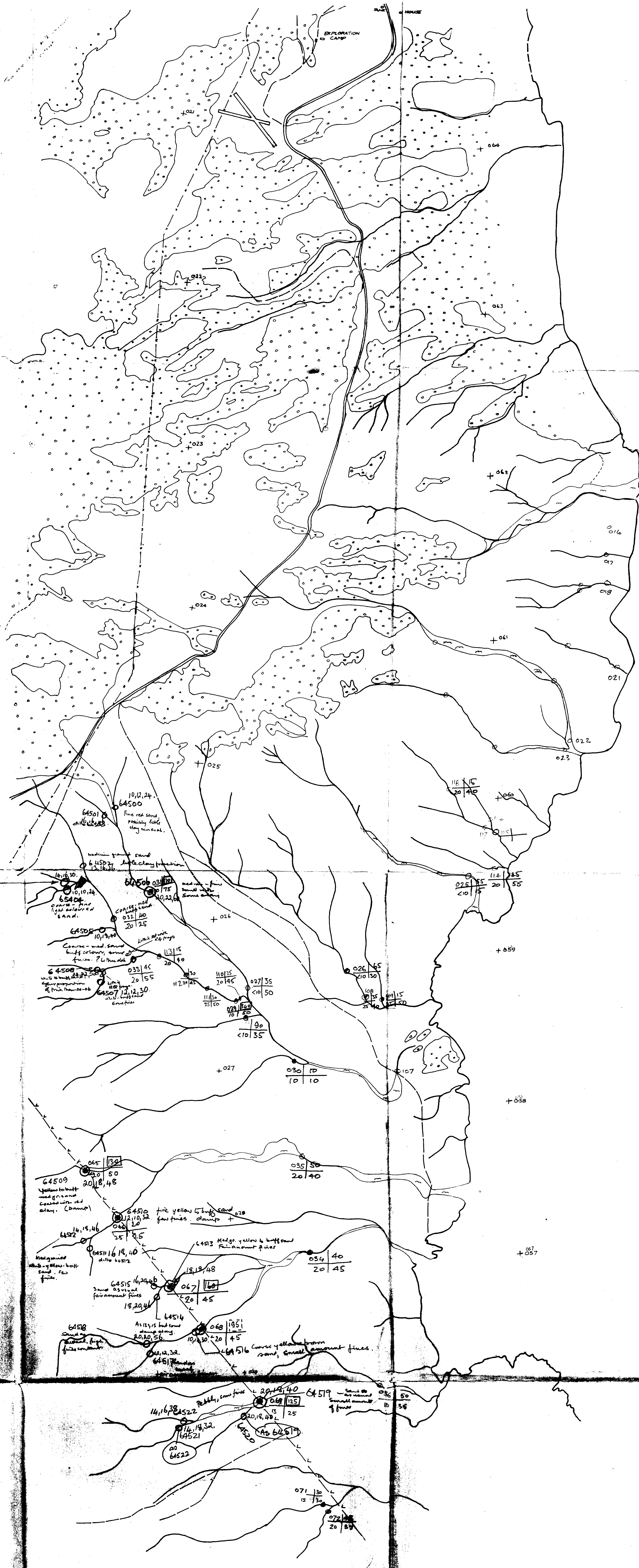
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[Symbol] Alluvium
 [Symbol] Sand dunes & Ridges

Sample No. Cu Drainage sample, values ppm (J. Christen) July 1971.
 Pb Zn
 + 038 Photo centres
 Roads
 Tracks
 Powerline
 Water line
 Creeks
 FOLLOW UP DRAINAGE SAMPLING
 OSHAID NO. 10 PLOT J. TAYLOR Sept. 1971
 Based on uncorrected airphoto mosaic

MOUNTAIN VIEW DISTRICT
 SML 522
 HOUSE TO MANGAROS WORKINGS
 DRAINAGE SAMPLING
 SCALE: 1:7500
 DATE: J.T. J.M. DATE:

6626-4



MOUNT GUNSON MINES PTY. LTD.		
SCALE: 1:5000	GEOLOGY: P.C. K.M.	DWG No
SURVEYED: R.K.Y.	DRAWN: R.K.Y.	
	CHECKED: R.K.Y.	
	DATE: 17/5/72	

6626-5



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